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(54) **DETECTING AN OPERATIONAL STATE OF A MEDIA DEVICE**

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H04N 21/4363 (2011.01)

(52) **U.S. Cl.**
CPC ... **H04N 21/4424** (2013.01); **H04N 21/43635** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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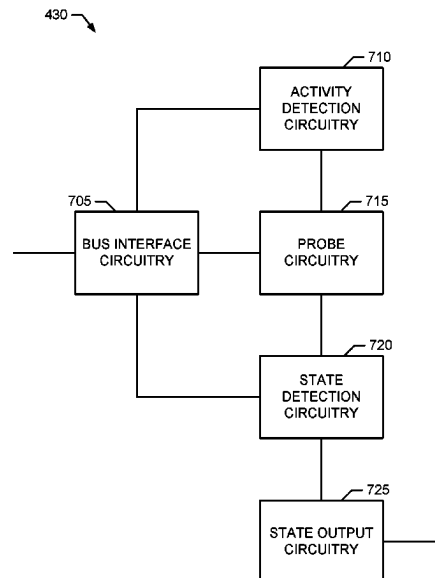
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(57) **ABSTRACT**

Example methods, apparatus, systems and articles of manufacture (e.g., physical storage media) to detect an operational state (such as an on/off state) of a media device are disclosed herein. Disclosed example apparatus are to monitor for activity on a first bus of an (HDMI) port of the media device. Disclosed example apparatus are also to inject a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration. Disclosed example apparatus are further to detect the operational state of the media device based on whether a response to the message is received on the first bus.

24 Claims, 11 Drawing Sheets



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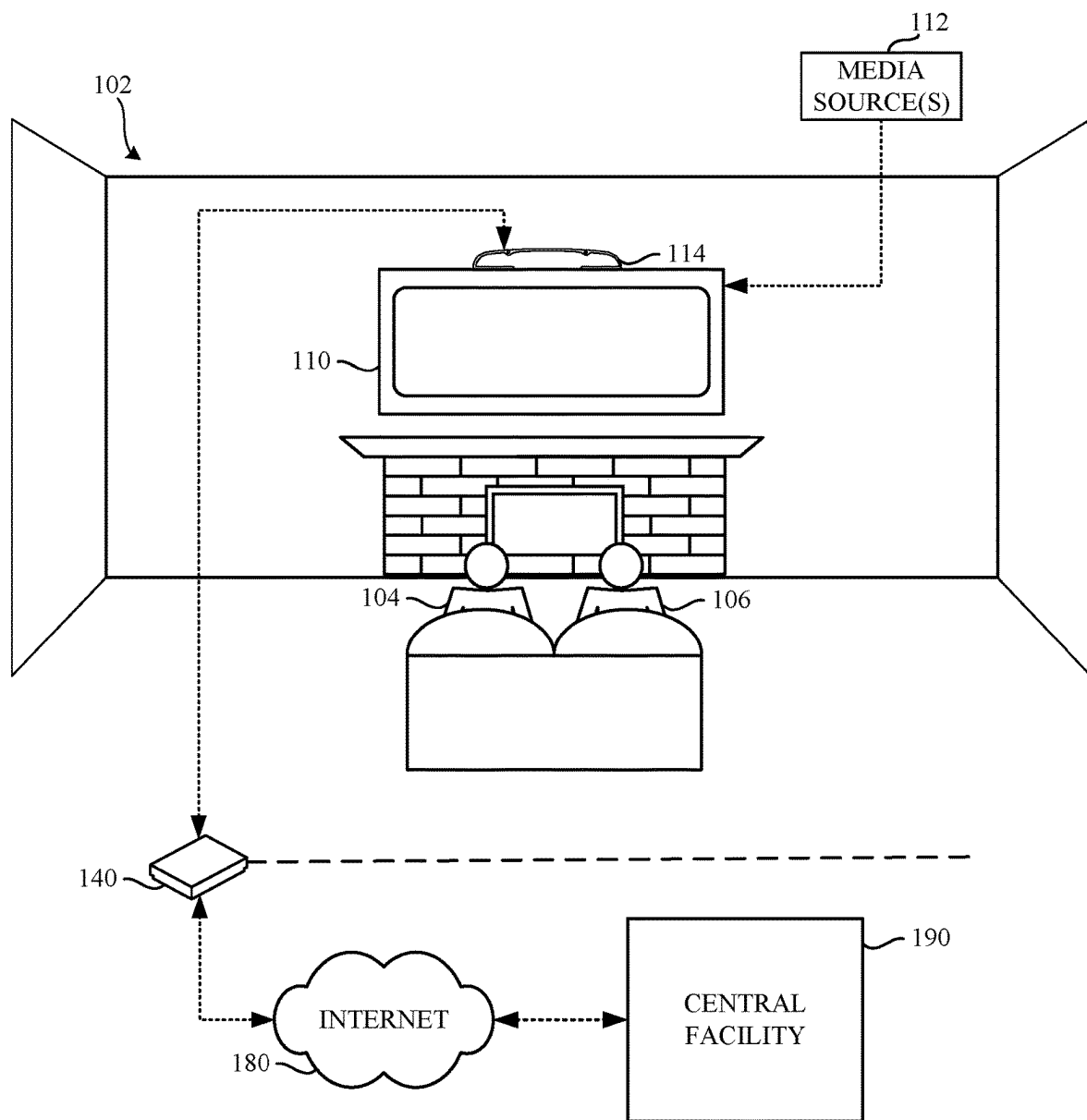


FIG. 1

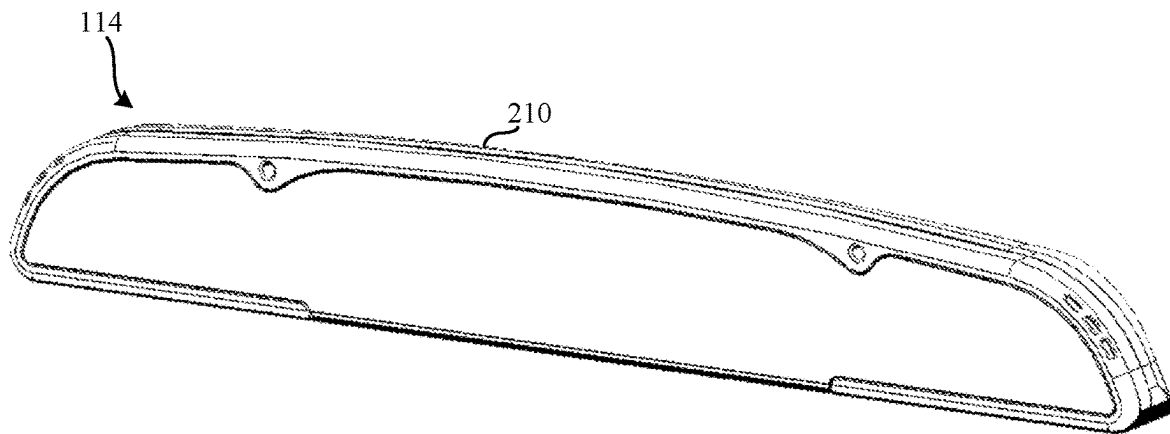


FIG. 2

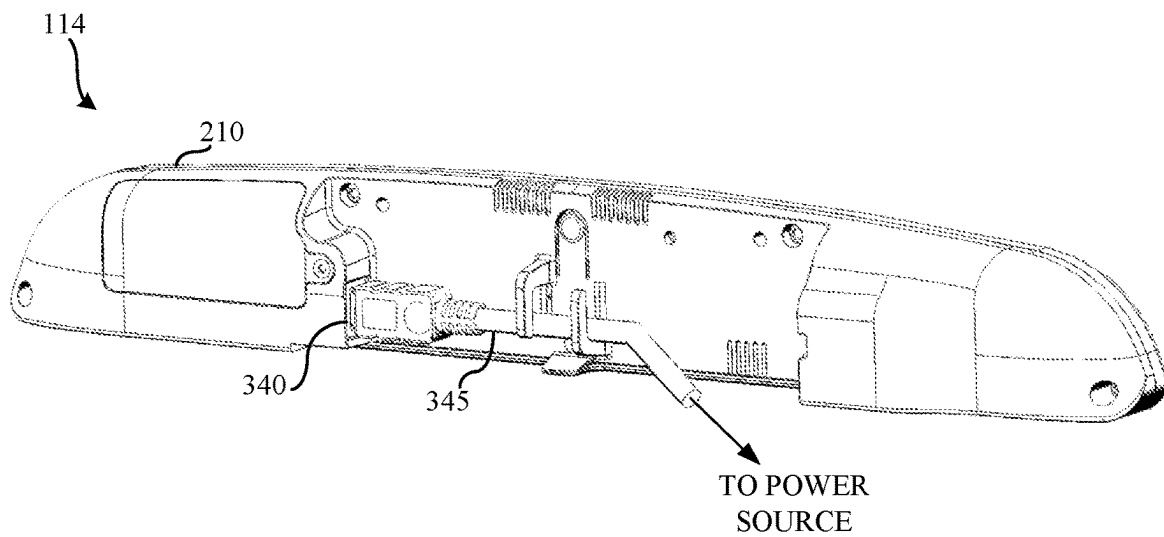


FIG. 3

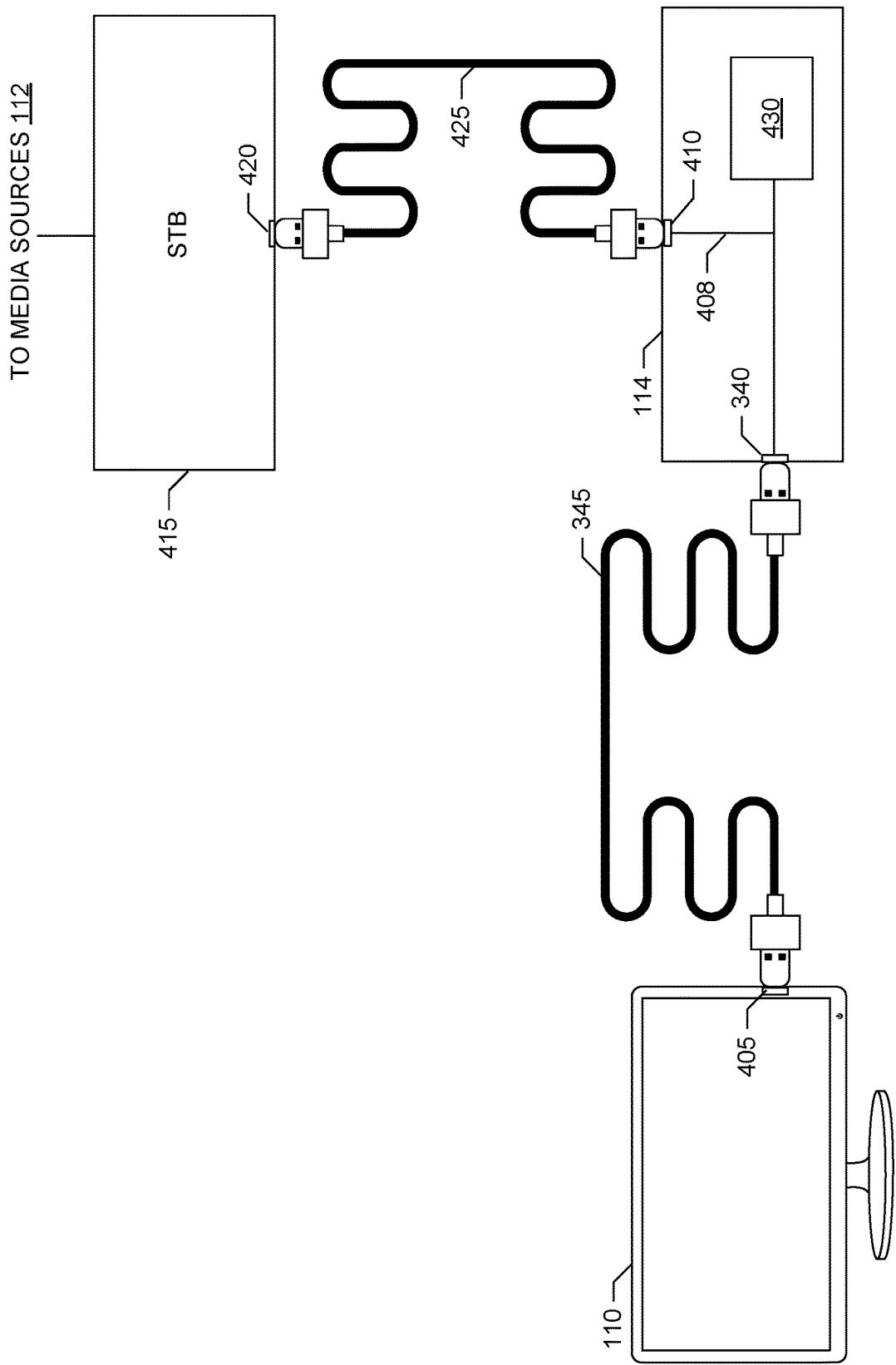


FIG. 4

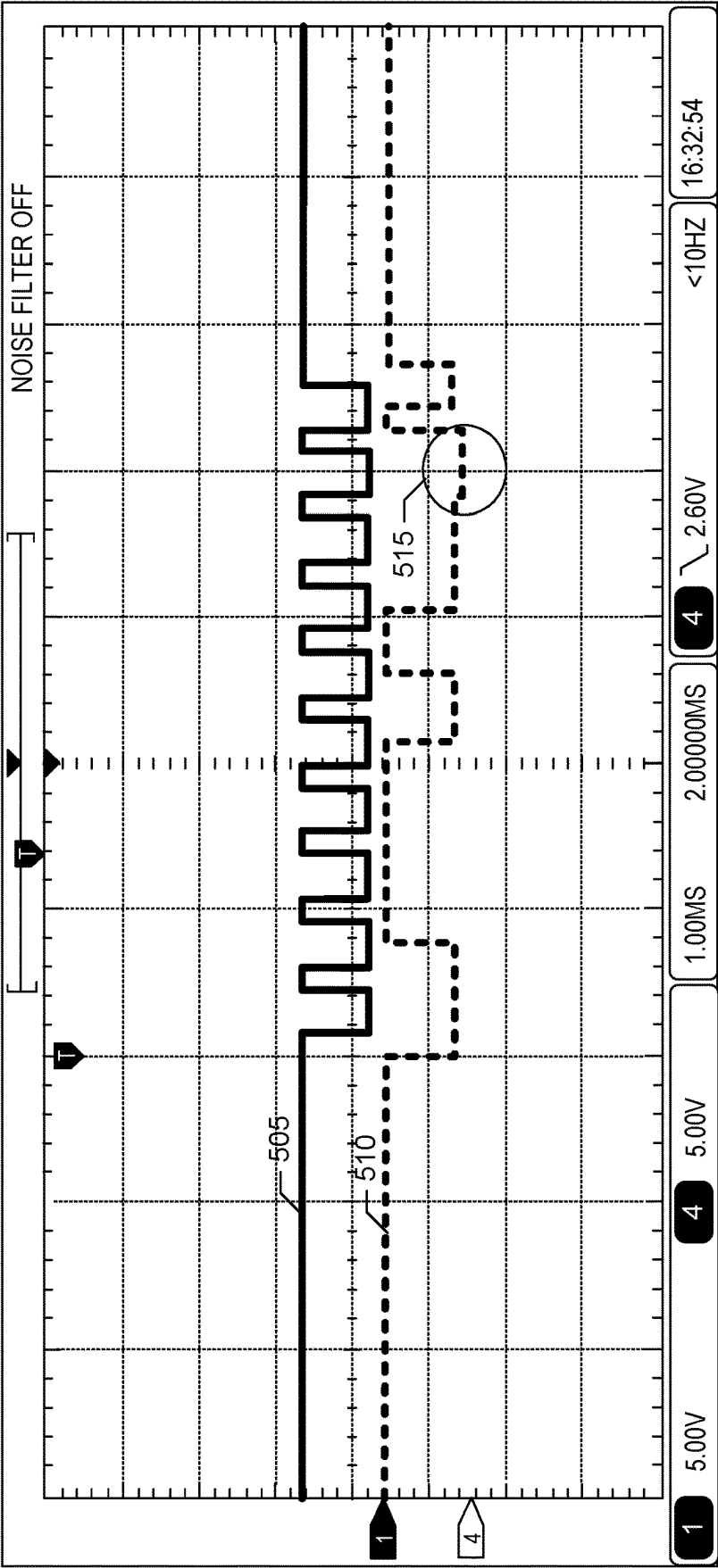


FIG. 5

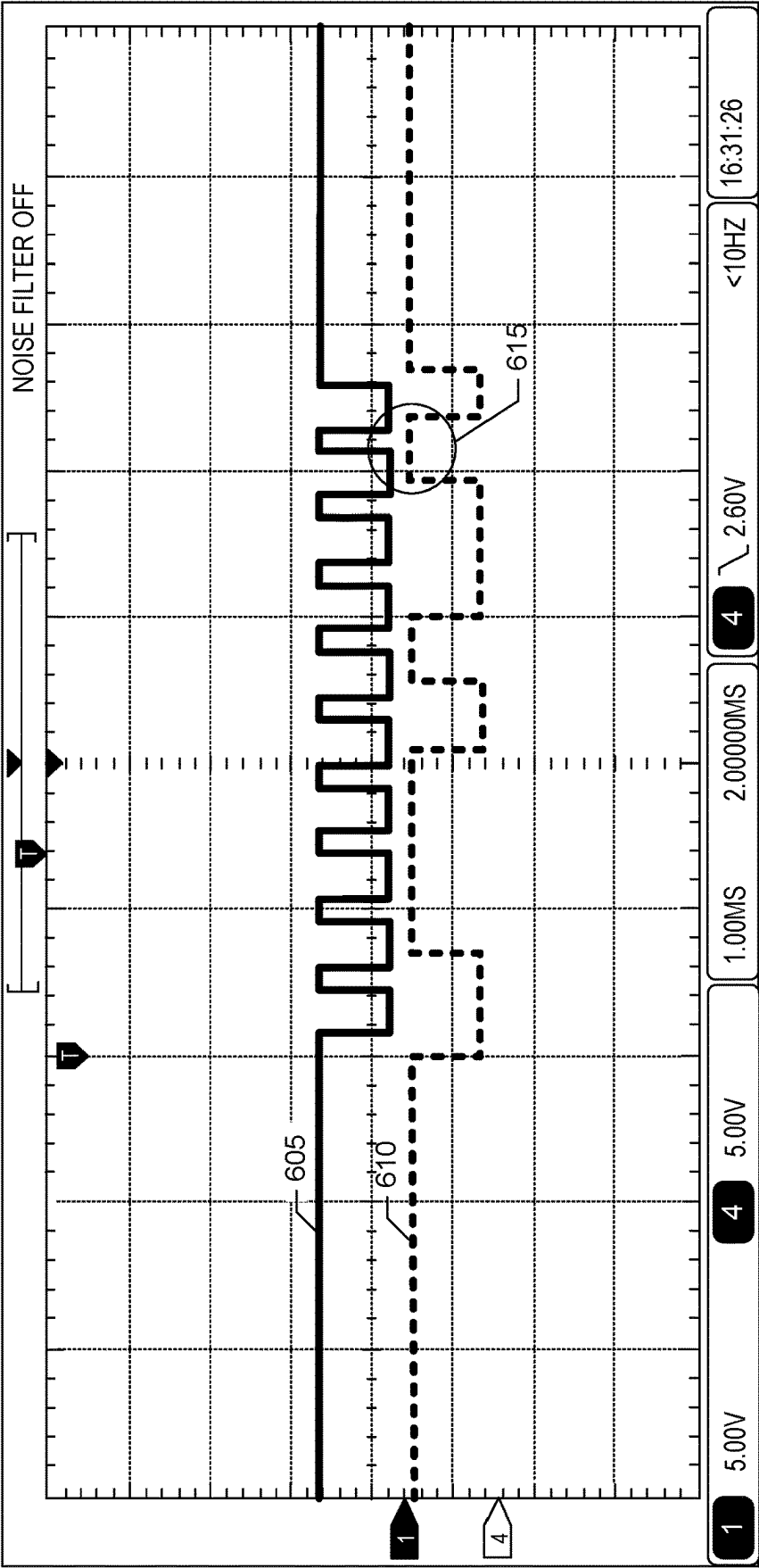


FIG. 6

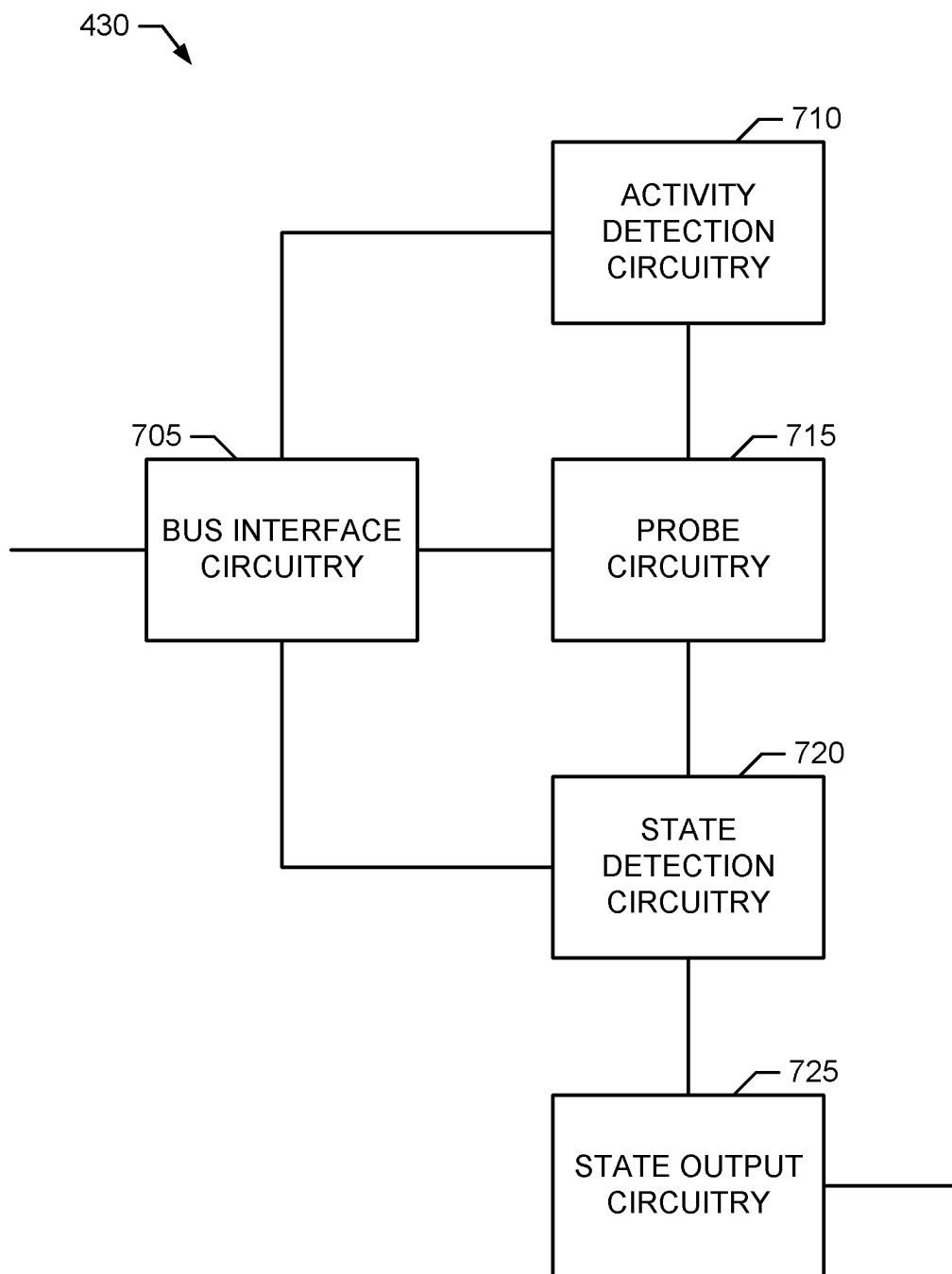


FIG. 7

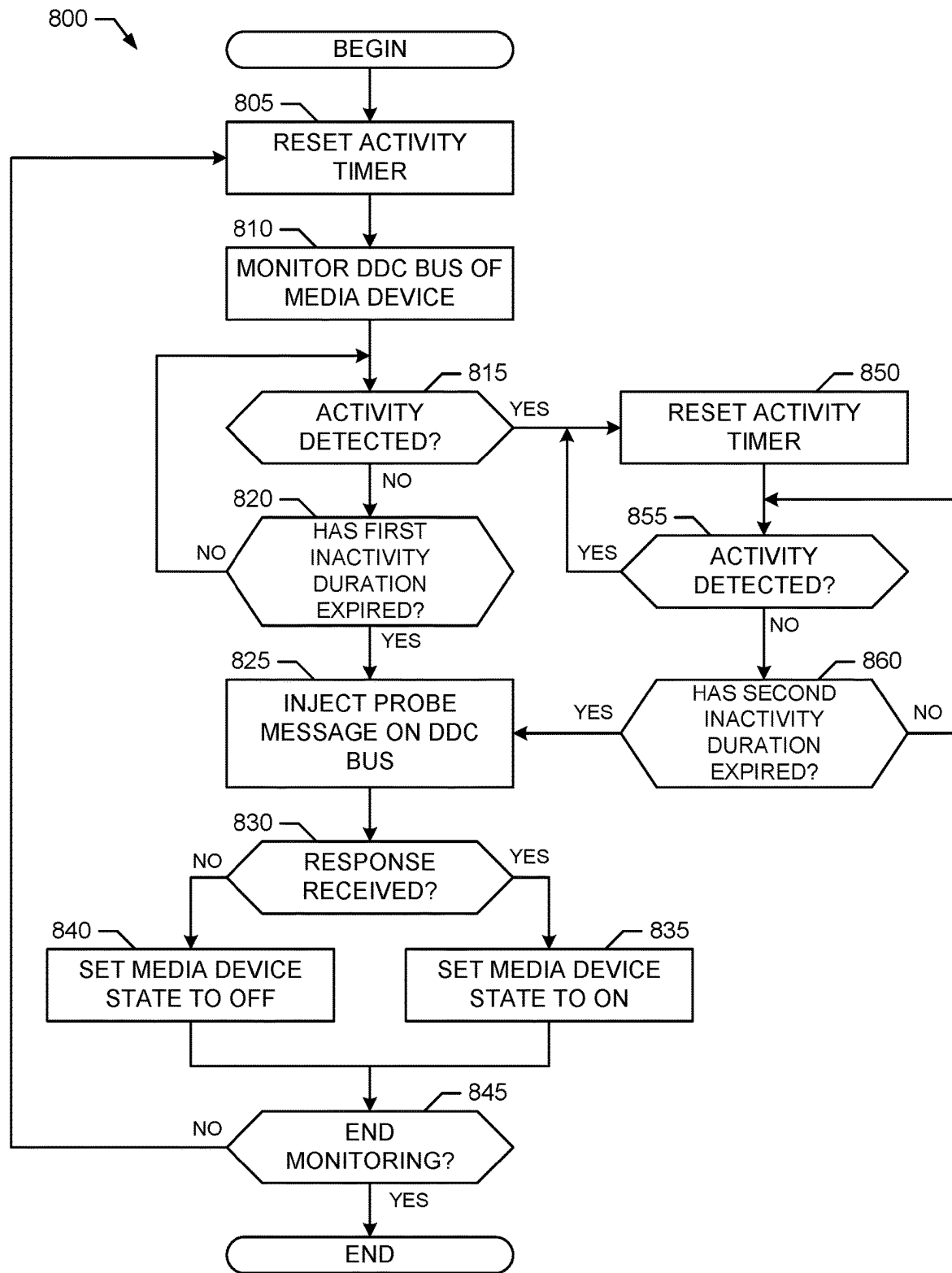


FIG. 8

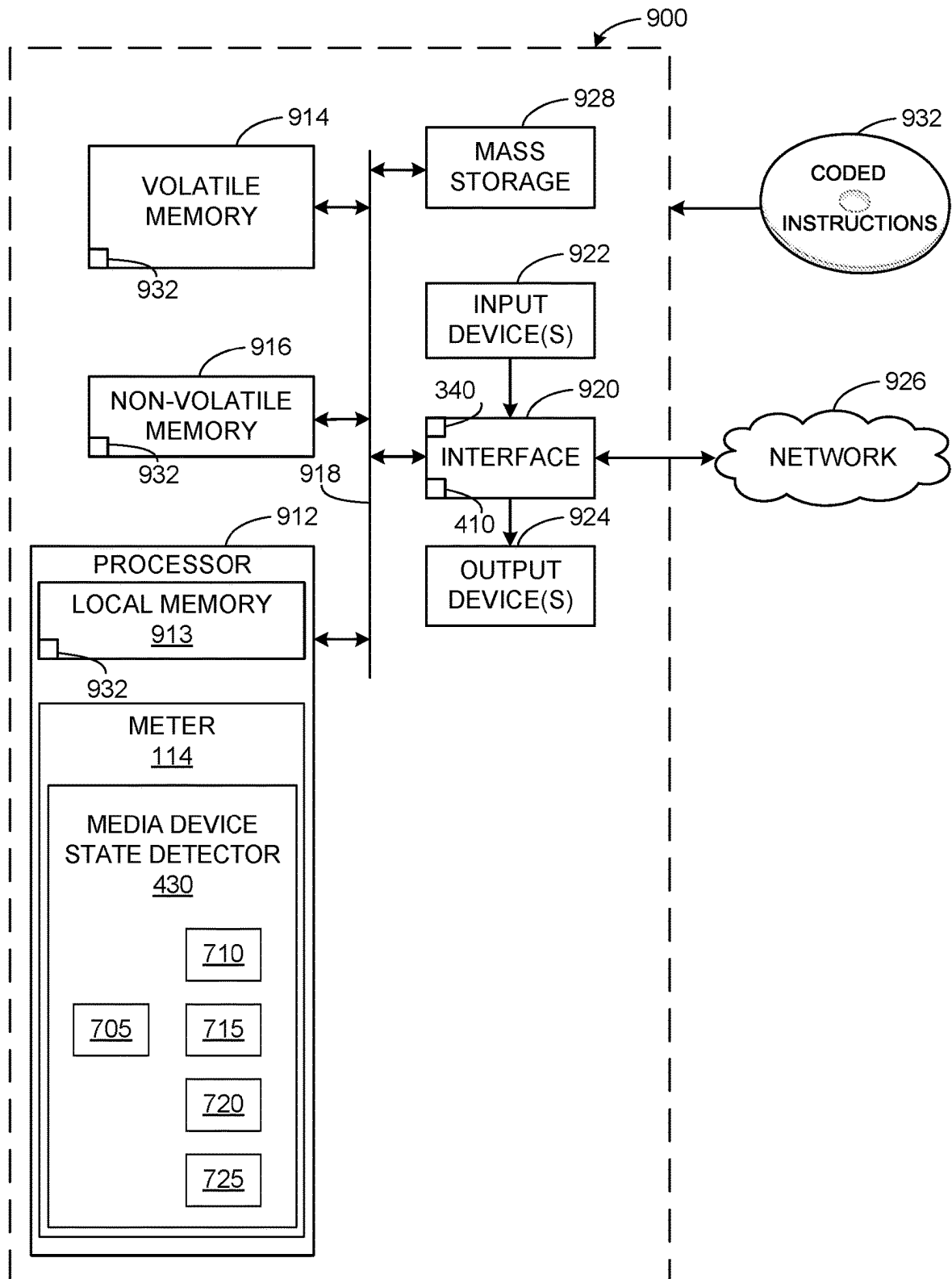


FIG. 9

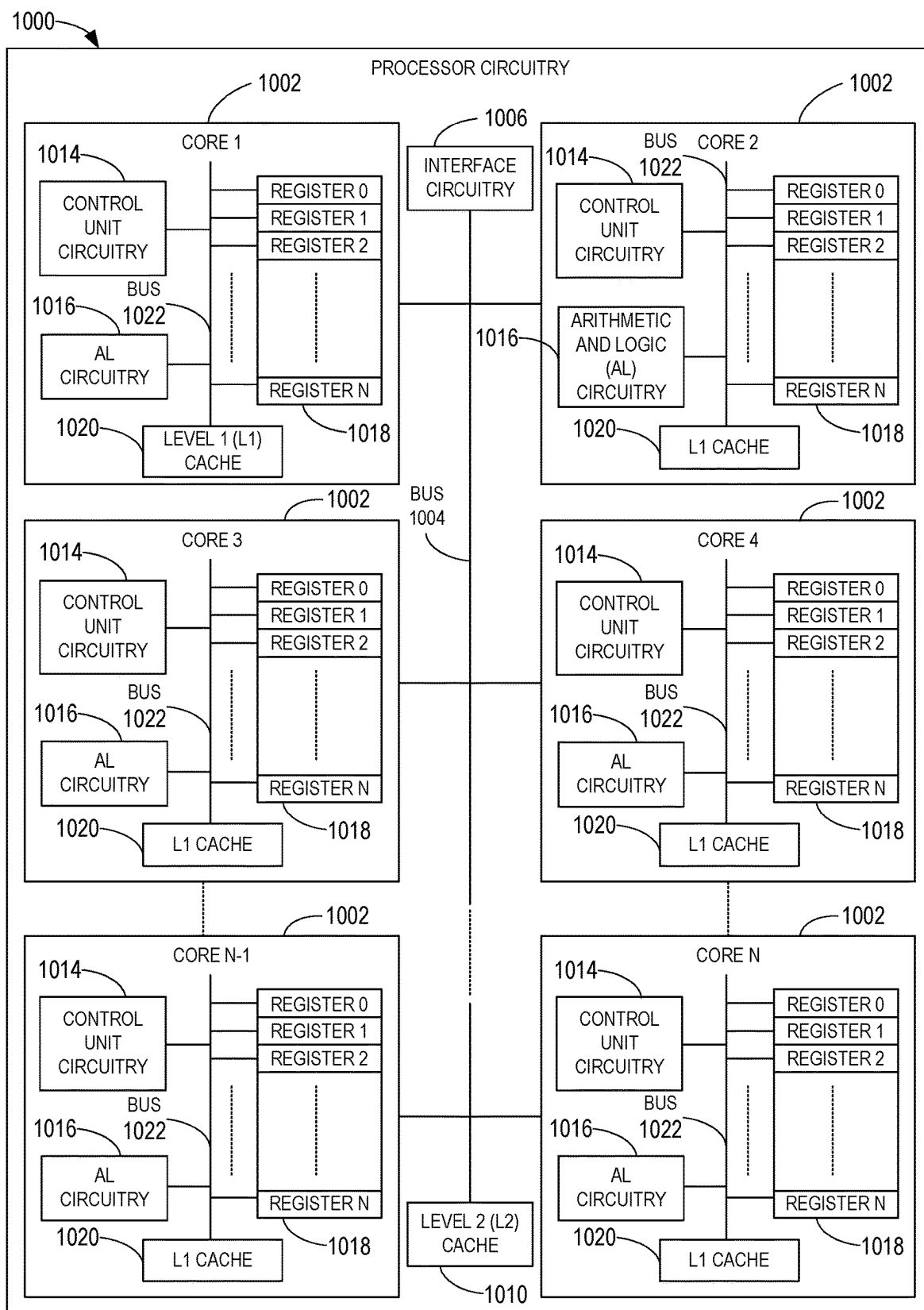


FIG. 10

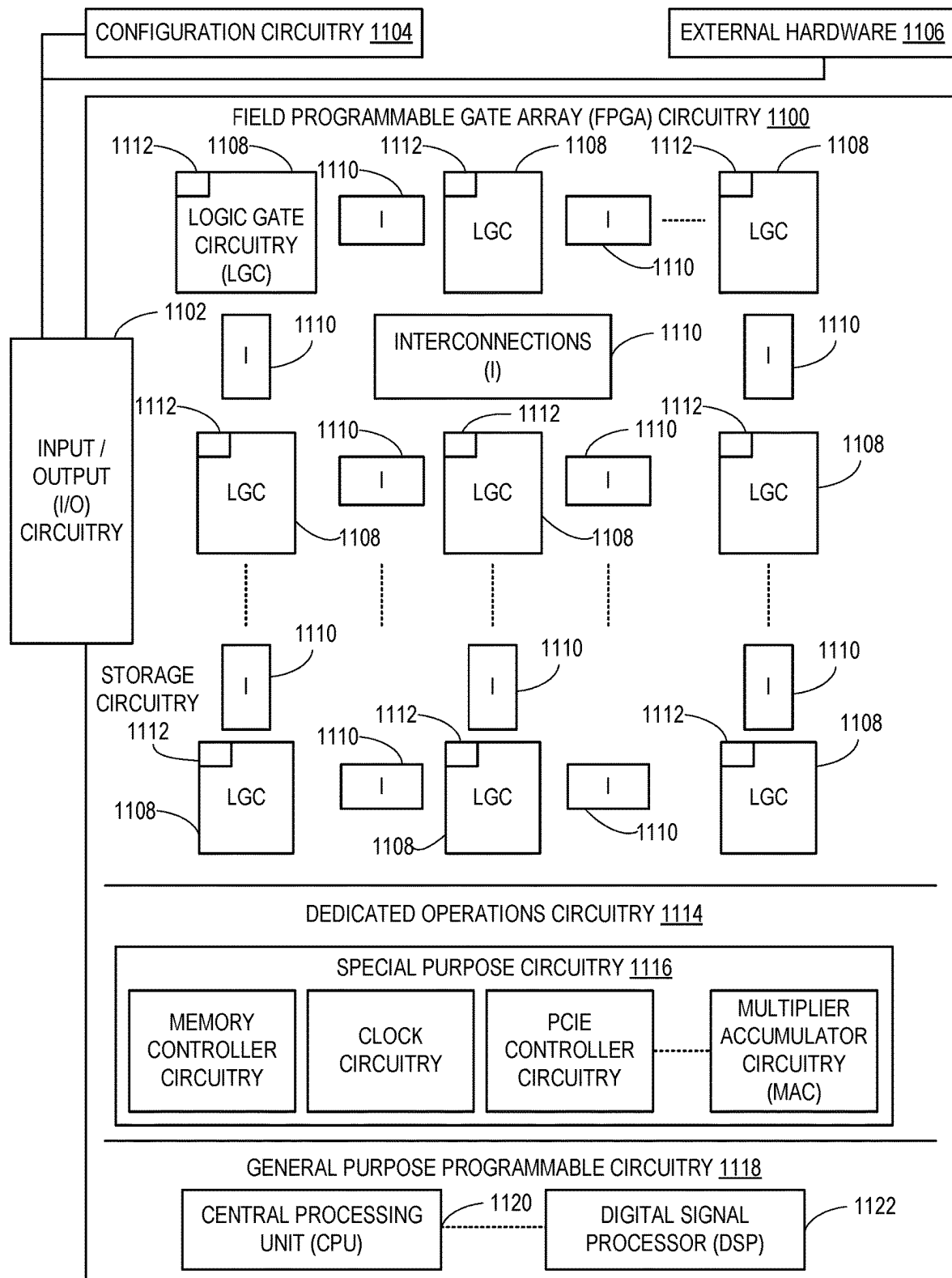


FIG. 11

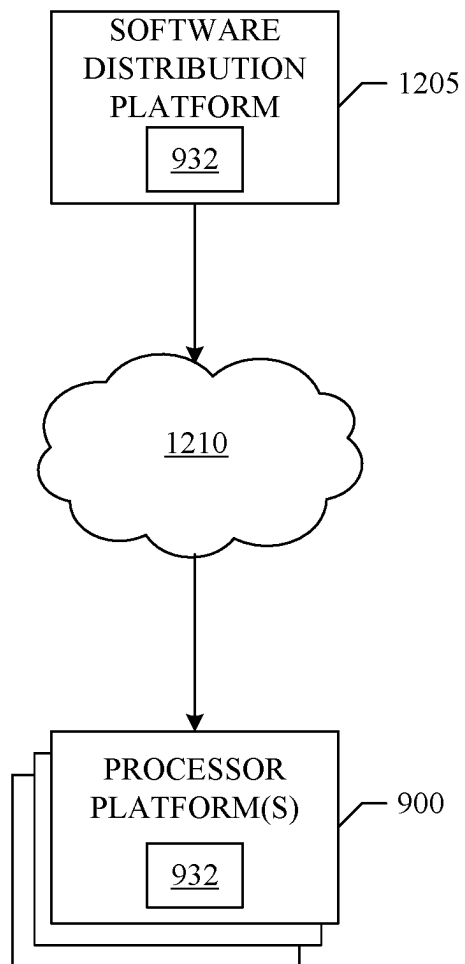


FIG. 12

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DETECTING AN OPERATIONAL STATE OF A MEDIA DEVICE

CROSS REFERENCE TO RELATED APPLICATION(S)

This patent arises from a continuation of U.S. patent application Ser. No. 18/075,126, now U.S. Pat. No. 11,838,585, which is titled “DETECTING AN OPERATIONAL STATE OF A MEDIA DEVICE”, and which was filed on Dec. 5, 2022, which is a continuation of U.S. patent application Ser. No. 17/529,792, now U.S. Pat. No. 11,523,178, which is titled “DETECTING AN OPERATIONAL STATE OF A MEDIA DEVICE,” and which was filed on Nov. 18, 2021, which claims the benefit of U.S. Provisional Application No. 63/116,620, which is titled “DETECTING ON/OFF STATE OF A MEDIA DEVICE,” and which was filed on Nov. 20, 2020. Priority to U.S. patent application Ser. No. 18/075,126, U.S. patent application Ser. No. 17/529,792 and U.S. Provisional Application No. 63/116,620 are claimed. U.S. patent application Ser. No. 18/075,126, U.S. patent application Ser. No. 17/529,792 and U.S. Provisional Application No. 63/116,620 are hereby incorporated by reference in their respective entireties.

FIELD OF THE DISCLOSURE

This disclosure relates generally to media device monitoring and, more particularly, to detecting an operational state of a media device.

BACKGROUND

Audience measurement systems typically include one or more site meters to monitor the media presented by one or more media devices located at a monitored site. In some arrangements, the monitored media device may receive media from one or more media sources, such as, but not limited to, a set-top box (STB), a digital versatile disk (DVD) player, a Blu-ray Disk™ player, a gaming console, a computer, etc., which are powered independently from the monitored media device. Accordingly, there is the possibility that, although a media source at the monitored site is powered on and providing media to the monitored media device, the monitored media device may be powered off and, thus, not actively presenting the media provided by the media source. Therefore, to enable accurate crediting of media exposure at the monitored site, some site meters further monitor the operational state of the monitored media device to determine whether the media device is powered off and not capable of presenting media, or powered on and capable of presenting media.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of an example system including an example device meter constructed to detect an operational state of a media device in accordance with the teachings of this disclosure.

FIG. 2 is an example front view of the example meter of FIG. 1.

FIG. 3 is an example rear view of the example meter of FIG. 1.

FIG. 4 is a block diagram of an example implementation of the meter of FIG. 1, which includes an example media device state detector implemented in accordance with the teachings of this disclosure.

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FIGS. 5-6 illustrate example signals monitored by the media device state detector of FIG. 4.

FIG. 7 is a block diagram of an example implementation of the media device state detector of FIG. 4.

FIG. 8 is a flowchart representative of example machine readable instructions and/or example operations that may be executed by example processor circuitry to implement the example media device state detector of FIGS. 4 and/or 7.

FIG. 9 is a block diagram of an example processor platform including processor circuitry structured to execute the example machine readable instructions and/or the example operations of FIG. 8 to implement the example media device state detector of FIGS. 4 and/or 7.

FIG. 10 is a block diagram of an example implementation of the processor circuitry of FIG. 9.

FIG. 11 is a block diagram of another example implementation of the processor circuitry of FIG. 9.

FIG. 12 is a block diagram of an example software distribution platform (e.g., one or more servers) to distribute software (e.g., software corresponding to the example machine readable instructions of FIG. 8) to client devices associated with end users and/or consumers (e.g., for license, sale and/or use), retailers (e.g., for sale, re-sale, license, and/or sub-license), and/or original equipment manufacturers (OEMs) (e.g., for inclusion in products to be distributed to, for example, retailers and/or to other end users such as direct buy customers).

In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts, elements, etc. The figures are not to scale. As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.

Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc. are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name. As used herein, “approximately” and “about” refer to dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections. As used herein “substantially real time” refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, “substantially real time” refers to real time+/-1 second.

As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but

rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

As used herein, “processor circuitry” is defined to include (i) one or more special purpose electrical circuits structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmed with instructions to perform specific operations and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of processor circuitry include programmed microprocessors, Field Programmable Gate Arrays (FPGAs) that may instantiate instructions, Central Processor Units (CPUs), Graphics Processor Units (GPUs), Digital Signal Processors (DSPs), XPU, or microcontrollers and integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of processor circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more DSPs, etc., and/or a combination thereof) and application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of the processing circuitry is/are best suited to execute the computing task(s).

DETAILED DESCRIPTION

Example methods, apparatus, systems and articles of manufacture (e.g., physical storage media) to detect an operational state (such as an on/off state) of a media device are disclosed herein. Modern media devices, such as flat-screen televisions (TVs), have very low power consumption. As such, conventional media device state detection techniques that are based on differentiation of the different power states associated with respective different media device (e.g., a monitored TV) states are becoming more and more difficult to calibrate and, thus, are potentially unreliable. High-definition multimedia interface (HDMI) ports are commonly available on modern TVs and other media devices, but the level of integration of HDMI features across media devices (e.g., TVs) from different manufacturers, and/or across different models of the same manufacturer, is inconsistent. A typical HDMI port includes several data buses, such as consumer electronics control (CEC) and display data channel (DDC) busses, with associated logical layer protocols that provide software commands that can report media device (e.g., TV) state. However, because the HDMI standard has been evolving over many years, commercially available media devices (e.g., TVs) have different levels of integration of the CEC and DDC specifications. This means that a media device state detection technique utilizing HDMI port monitoring may be unable to rely on, for example, a monitored TV answering a “TV State” CEC query signaled over the HDMI port, as that command might not have been implemented yet in the HDMI protocol stack of the particular monitored TV. Therefore, CEC and DDC protocols may be unreliable for media device state detection, and/or at least difficult to maintain.

Example media device state detection techniques disclosed herein utilize an HDMI port of a monitored media device (e.g., TV) to detect an operational state of the monitored media device, such as whether the monitored media device is on or off. However, disclosed example media device state detection techniques do not rely on any

particular protocol commands being implemented over the HDMI port to detect the operational state of the monitored media device (e.g., TV). Rather, a disclosed example media device state detector monitors the DDC bus of the HDMI port for activity. When the media device state detector finds a gap in activity of at least a threshold duration, the media device state detector sends a probe message (also referred to as a probe data frame) on the DDC bus. If the media device (e.g., the TV) is switched on (is in the on operational state) the inter-integrated circuit (I2C) electrical subsystem that implements the DDC bus will acknowledge the probe message regardless of whether the upper layer DDC protocol is implemented or has recognized the command represented by the probe message. However, if the media device (e.g., the TV) is switched off (is in the off operational state), the probe message will not be acknowledged by the I2C electrical subsystem. In example implementations disclosed herein, a media device user (e.g., a TV user) will not notice any consequences of the above media device state detection technique because the media device state detector implements a communication collision avoidance algorithm, as disclosed herein, which avoids communication collisions with other devices present on the DDC bus.

These and other example methods, apparatus, systems and articles of manufacture (e.g., physical storage media) to detect an operational state of a media device are disclosed in further detail below.

FIG. 1 is an illustration of an example audience measurement system constructed to include functionality to detect the operational state (e.g., the on/off state) of a media device in accordance with the teachings of this disclosure. In the illustrated example of FIG. 1, an example media presentation environment 102 includes example panelists 104, 106, an example media device 110 (also referred to as a media presentation device) that receives media from an example media source 112, and an example meter 114. The example meter 114 identifies the media presented by the example media device 110 and reports media monitoring information to an example central facility 190 of an example audience measurement entity via an example gateway 140 and an example network 180. In some examples, the meter 114 is referred to as a site meter, a device meter, an audience measurement device, etc. As disclosed in further detail below, the meter 114 is able to detect the operational state (e.g., the on/off state) of the media device 110 in accordance with the teachings of this disclosure.

In the illustrated example of FIG. 1, the example media presentation environment 102 is a room of a household (e.g., a room in a home of a panelist, such as the home of a “Nielsen family”). In the illustrated example of FIG. 1, the example panelists 104, 106 of the household have been statistically selected to develop media ratings data (e.g., television ratings data) for a population/demographic of interest. People become panelists via, for example, a user interface presented on a media device (e.g., via the media device 110, via a website, etc.). People become panelists in additional or alternative manners such as, for example, via a telephone interview, by completing an online survey, etc. Additionally or alternatively, people may be contacted and/or enlisted using any desired methodology (e.g., random selection, statistical selection, phone solicitations, Internet advertisements, surveys, advertisements in shopping malls, product packaging, etc.). In some examples, an entire family may be enrolled as a household of panelists. That is, while a mother, a father, a son, and a daughter may each be identified as individual panelists, their viewing activities typically occur within the family’s household.

In the illustrated example of FIG. 1, one or more panelists **104**, **106** of the household have registered with an audience measurement entity (e.g., by agreeing to be a panelist) and have provided their demographic information to the audience measurement entity as part of a registration process to enable associating demographics with media exposure activities (e.g., television exposure, radio exposure, Internet exposure, etc.). The demographic data includes, for example, age, gender, income level, educational level, marital status, geographic location, race, etc., of a panelist. While the example media presentation environment **102** is a household in the illustrated example of FIG. 1, the example media presentation environment **102** can additionally or alternatively be any other type(s) of environments such as, for example, a theater, a restaurant, a tavern, a retail location, an arena, etc.

In the illustrated example of FIG. 1, the example media device **110** is a television. However, the example media device **110** can correspond to any type of audio, video and/or multimedia device capable of presenting media audibly and/or visually. In some examples, the media device **110** (e.g., a television) may communicate audio to another media device (e.g., an audio/video receiver) for output by one or more speakers (e.g., surround sound speakers, a sound bar, etc.). As another example, the media device **110** can correspond to a multimedia computer system, a personal digital assistant, a cellular/mobile smartphone, a radio, a home theater system, stored audio and/or video played back from a memory, such as a digital video recorder or a digital versatile disc, a webpage, and/or any other communication device capable of presenting media to an audience (e.g., the panelists **104**, **106**).

The media device **110** receives media from the media source **112**. The media source **112** may be any type of media provider(s), such as, but not limited to, a cable media service provider, a radio frequency (RF) media provider, an Internet based provider (e.g., IPTV), a satellite media service provider, etc., and/or any combination thereof. The media may be radio media, television media, pay per view media, movies, Internet Protocol Television (IPTV), satellite television (TV), Internet radio, satellite radio, digital television, digital radio, stored media (e.g., a compact disk (CD), a Digital Versatile Disk (DVD), a Blu-ray disk, etc.), any other type(s) of broadcast, multicast and/or unicast medium, audio and/or video media presented (e.g., streamed) via the Internet, a video game, targeted broadcast, satellite broadcast, video on demand, etc. For example, the media device **110** can correspond to a television and/or display device that supports the National Television Standards Committee (NTSC) standard, the Phase Alternating Line (PAL) standard, the Système Électronique pour Couleur avec Mémoire (SECAM) standard, a standard developed by the Advanced Television Systems Committee (ATSC), such as high definition television (HDTV), a standard developed by the Digital Video Broadcasting (DVB) Project, etc. Advertising, such as an advertisement and/or a preview of other programming that is or will be offered by the media source **112**, etc., is also typically included in the media.

In examples disclosed herein, an audience measurement entity provides the meter **114** to the panelist **104**, **106** (or household of panelists). The meter **114** may be installed by the panelist **104**, **106** by simply powering the meter **114** and placing the meter **114** in the media presentation environment **102** and/or near the media device **110** (e.g., near a television set). In some examples, more complex installation activities may be performed such as, for example, affixing the meter **114** to the media device **110**, electronically connecting the

meter **114** to the media device **110**, etc. The example meter **114** detects exposure to media and electronically stores monitoring information (e.g., a code detected with the presented media, a signature of the presented media, an identifier of a panelist that is present at the time of the presentation, a timestamp of the time of the presentation, etc.) of the presented media. The stored monitoring information is then transmitted back to the central facility **190** via the gateway **140** and the network **180**. While the media monitoring information is transmitted by electronic transmission in the illustrated example of FIG. 1, the media monitoring information may additionally or alternatively be transferred in any other manner, such as, for example, by physically mailing the meter **114**, by physically mailing a memory of the meter **114**, etc.

The meter **114** of the illustrated example combines audience measurement data and audience identification data. For example, audience measurement data is determined by monitoring media output by the media device **110** and/or other media device(s), and audience identification data (also referred to as demographic data, people monitoring data, etc.) is determined from people monitoring data provided to the meter **114**. Thus, the example meter **114** provides dual functionality of an audience measurement meter that is to collect audience measurement data, and a people meter that is to collect and/or associate demographic information corresponding to the collected audience measurement data.

For example, the meter **114** of the illustrated example collects media identifying information and/or data (e.g., signature(s), fingerprint(s), code(s), tuned channel identification information, time of exposure information, etc.) and people monitoring data (e.g., user identifiers, demographic data associated with audience members, etc.). The media identifying information and the people monitoring data can be combined to generate, for example, media exposure data (e.g., ratings data) indicative of amount(s) and/or type(s) of people that were exposed to specific piece(s) of media distributed via the media device **110**. To extract media identification data, the meter **114** of the illustrated example of FIG. 1 monitors for watermarks (sometimes referred to as codes) included in the presented media and/or generates signatures (sometimes referred to as fingerprints) representative of the presented media.

Audio watermarking is a technique used to identify media such as television broadcasts, radio broadcasts, advertisements (television and/or radio), downloaded media, streaming media, prepackaged media, etc. Existing audio watermarking techniques identify media by embedding one or more audio codes (e.g., one or more watermarks), such as media identifying information and/or an identifier that may be mapped to media identifying information, into an audio and/or video component. In some examples, the audio or video component is selected to have a signal characteristic sufficient to hide the watermark. As used herein, the terms “code” or “watermark” are used interchangeably and are defined to mean any identification information (e.g., an identifier) that may be inserted or embedded in the audio or video of media (e.g., a program or advertisement) for the purpose of identifying the media or for another purpose such as tuning (e.g., a packet identifying header). As used herein “media” refers to audio and/or visual (still or moving) content and/or advertisements. To identify watermarked media, the watermark(s) are extracted and used to access a table of reference watermarks that are mapped to media identifying information.

Unlike media monitoring techniques based on codes and/or watermarks included with and/or embedded in the

monitored media, fingerprint or signature-based media monitoring techniques generally use one or more inherent characteristics of the monitored media during a monitoring time interval to generate a substantially unique proxy for the media. Such a proxy is referred to as a signature or fingerprint, and can take any form (e.g., a series of digital values, a waveform, etc.) representative of any aspect(s) of the media signal(s) (e.g., the audio and/or video signals forming the media presentation being monitored). A signature may be a series of signatures collected in series over a timer interval. A good signature is repeatable when processing the same media presentation, but is unique relative to other (e.g., different) presentations of other (e.g., different) media. Accordingly, the term “fingerprint” and “signature” are used interchangeably herein and are defined herein to mean a proxy for identifying media that is generated from one or more inherent characteristics of the media.

Signature-based media monitoring generally involves determining (e.g., generating and/or collecting) signature(s) representative of a media signal (e.g., an audio signal and/or a video signal) output by a monitored media device and comparing the monitored signature(s) to one or more reference signatures corresponding to known (e.g., reference) media sources. Various comparison criteria, such as a cross-correlation value, a Hamming distance, etc., can be evaluated to determine whether a monitored signature matches a particular reference signature. When a match between the monitored signature and one of the reference signatures is found, the monitored media can be identified as corresponding to the particular reference media represented by the reference signature that with matched the monitored signature. Because attributes, such as an identifier of the media, a presentation time, a broadcast channel, etc., are collected for the reference signature, these attributes may then be associated with the monitored media whose monitored signature matched the reference signature. Example systems for identifying media based on codes and/or signatures are long known and were first disclosed in Thomas, U.S. Pat. No. 5,481,294, which is hereby incorporated by reference in its entirety.

Depending on the type(s) of metering the meter **114** is to perform, the meter **114** can be physically coupled to the media device **110** or may be configured to capture audio emitted externally by the media device **110** (e.g., free field audio) such that direct physical coupling to the media device **110** is not required. For example, the meter **114** of the illustrated example may employ non-invasive monitoring not involving any physical connection to the media device **110** (e.g., via Bluetooth® connection, WIFI® connection, acoustic sensing via one or more microphone(s) and/or other acoustic sensor(s), etc.) and/or invasive monitoring involving one or more physical connections to the media device **110** (e.g., via USB connection, a High Definition Media Interface (HDMI) connection, an Ethernet cable connection, etc.).

In examples disclosed herein, to monitor media presented by the media device **110**, the meter **114** of the illustrated example senses audio (e.g., acoustic signals or ambient audio) output (e.g., emitted) by the media device **110**. For example, the meter **114** processes the signals obtained from the media device **110** to detect media and/or source identifying signals (e.g., audio watermarks, audio signatures) embedded in and/or generated from portion(s) (e.g., audio portions) of the media presented by the media device **110**. To, for example, sense ambient audio output by the media device **110**, the meter **114** of the illustrated example includes an example acoustic sensor (e.g., a microphone). In some

examples, the meter **114** may process audio signals obtained from the media device **110** via a direct cable connection to detect media and/or source identifying audio watermarks embedded in such audio signals.

To generate exposure data for the media, identification(s) of media to which the audience is exposed are correlated with people data (e.g., presence information) collected by the meter **114**. The meter **114** of the illustrated example collects inputs (e.g., audience identification data) representative of the identities of the audience member(s) (e.g., the panelists **104**, **106**). In some examples, the meter **114** collects audience identification data by periodically and/or a-periodically prompting audience members in the media presentation environment **102** to identify themselves as present in the audience. In some examples, the meter **114** responds to predetermined events (e.g., when the media device **110** is turned on, a channel is changed, an infrared control signal is detected, etc.) by prompting the audience member(s) to self-identify. The audience identification data and the exposure data can then be complied with the demographic data collected from audience members such as, for example, the panelists **104**, **106** during registration to develop metrics reflecting, for example, the demographic composition of the audience. The demographic data includes, for example, age, gender, income level, educational level, marital status, geographic location, race, etc., of the panelist.

In some examples, the meter **114** may be configured to receive panelist information via an input device such as, for example, a remote control, an Apple® iPad®, a cell phone, etc. In such examples, the meter **114** prompts the audience members to indicate their presence by pressing an appropriate input key on the input device. The meter **114** of the illustrated example may also determine times at which to prompt the audience members to enter information to the meter **114**. In some examples, the meter **114** of FIG. **1** supports audio watermarking for people monitoring, which enables the meter **114** to detect the presence of a panelist-identifying metering device in the vicinity (e.g., in the media presentation environment **102**) of the media device **110**. For example, the acoustic sensor of the meter **114** is able to sense example audio output (e.g., emitted) by an example panelist-identifying metering device, such as, for example, a wristband, a cell phone, etc., that is uniquely associated with a particular panelist. The audio output by the example panelist-identifying metering device may include, for example, one or more audio watermarks to facilitate identification of the panelist-identifying metering device and/or the panelist **104** associated with the panelist-identifying metering device.

The meter **114** of the illustrated example communicates with a remotely located central facility **190** of the audience measurement entity. In the illustrated example of FIG. **1**, the example meter **114** communicates with the central facility **190** via a gateway **140** and a network **180**. The example meter **114** of FIG. **1** sends media identification data and/or audience identification data to the central facility **190** periodically, a-periodically and/or upon request by the central facility **190**.

The example gateway **140** of the illustrated example of FIG. **1** can be implemented by a router that enables the meter **114** and/or other devices in the media presentation environment (e.g., the media device **110**) to communicate with the network **180** (e.g., the Internet).

In some examples, the example gateway **140** facilitates delivery of media from the media source(s) **112** to the media device **110** via the Internet. In some examples, the example

gateway **140** includes gateway functionality such as modem capabilities. In some other examples, the example gateway **140** is implemented in two or more devices (e.g., a router, a modem, a switch, a firewall, etc.). The gateway **140** of the illustrated example may communicate with the network **126** via Ethernet, a digital subscriber line (DSL), a telephone line, a coaxial cable, a USB connection, a Bluetooth connection, any wireless connection, etc.

In some examples, the example gateway **140** hosts a Local Area Network (LAN) for the media presentation environment **102**. In the illustrated example, the LAN is a wireless local area network (WLAN), and allows the meter **114**, the media device **110**, etc., to transmit and/or receive data via the Internet. Alternatively, the gateway **140** may be coupled to such a LAN.

The network **180** of the illustrated example can be implemented by a wide area network (WAN) such as the Internet. However, in some examples, local networks may additionally or alternatively be used. Moreover, the example network **180** may be implemented using any type of public or private network such as, but not limited to, the Internet, a telephone network, a local area network (LAN), a cable network, and/or a wireless network, or any combination thereof.

The central facility **190** of the illustrated example is implemented by one or more servers. The central facility **190** processes and stores data received from the meter(s) **114**. For example, the example central facility **190** of FIG. 1 combines audience identification data and program identification data from multiple households to generate aggregated media monitoring information. The central facility **190** generates reports for advertisers, program producers and/or other interested parties based on the compiled statistical data. Such reports include extrapolations about the size and demographic composition of audiences of content, channels and/or advertisements based on the demographics and behavior of the monitored panelists.

As noted above, the meter **114** of the illustrated example provides a combination of media metering and people metering. The meter **114** of FIG. 1 includes its own housing, processor, memory and/or software to perform the desired media monitoring and/or people monitoring functions. The example meter **114** of FIG. 1 is a stationary device disposed on or near the media device **110**. To identify and/or confirm the presence of a panelist present in the media presentation environment **102**, the example meter **114** of the illustrated example includes a display. For example, the display provides identification of the panelists **104**, **106** present in the media presentation environment **102**. For example, in the illustrated example, the meter **114** displays indicia (e.g., illuminated numerical numerals **1**, **2**, **3**, etc.) identifying and/or confirming the presence of the first panelist **104**, the second panelist **106**, etc. In the illustrated example, the meter **114** is affixed to a top of the media device **110**. However, the meter **114** may be affixed to the media device in any other orientation, such as, for example, on a side of the media device **110**, on the bottom of the media device **110**, and/or may not be affixed to the media device **110**. For example, the meter **114** may be placed in a location near the media device **110**.

FIG. 2 is an example front view of the example meter **114** of FIG. 1. In the illustrated example of FIG. 2, the example meter **114** includes an example housing **210**. In examples disclosed herein, the housing **210** is to be affixed to the media device **110**. For example, the housing may be affixed to a top of the media device **110**, may be affixed to a bottom of the media device **110**, may be affixed to a side of the media device **110**, etc. In some examples, the housing **210**

of the meter **114** is not affixed to the media device **110**. For example, the housing **210** may be placed in any other location within the media presentation environment **102** such that audio may be received by the meter **114**.

FIG. 3 is an example rear view of the example meter **114** of FIG. 1. In the illustrated example of FIG. 3, the example housing **210** includes an example HDMI port **340**. In the illustrated example of FIG. 3, the HDMI port **340** enables connection of the example meter **114** to an HDMI port of the media device **110** via an HDMI cable **345**.

FIG. 4 is a block diagram of an example implementation of the meter **114** of FIGS. 1-3, and further illustrates an example of interconnecting the meter **114** with the example media device **110**. In the illustrated example of FIG. 4, the meter **114** is electrically connected to an example HDMI port **405** of the media device **110** via the HDMI port **340** of the meter **114** and the HDMI cable **345**. In the illustrated example of FIG. 4, the meter **114** provides an example pass-through electrical connection **408** to another example HDMI port **410** of the meter **114**. The port HDMI port **410** can be electrically connected to another media device, such as an example set-top box (STB) **415**, via an example HDMI port **420** of the STB **415** and another example HDMI cable **425**. In the illustrated example, STB **415** obtains media from one or more of the media sources **112**. By providing the pass-through connection **408**, the meter **114** can bridge the HDMI port **405** of media device **110** with the HDMI port **420** of the STB **415**, thereby enabling the STB **415** to communicate with the media device **110** via their respective HDMI ports **420** and **405**. Of course, the meter **114** is not limited to bridging the media device **110** with the STB **415**, but can bridge the media device **110** with any other device having an HDMI port. (In other words, the STB **415** of FIG. 4 can be replaced with any device having an HDMI port.)

In the illustrated example, the meter **114** also includes an example media device state detector **430** implemented in accordance with teaching of this disclosure. The media device state detector **430** monitors the DDC bus of the HDMI port **405** of the media device **110** by monitoring the corresponding DDC bus pins of the pass-through connection **408**. As described above, the media device state detector **430** monitors the DDC bus of the HDMI port **405** for activity. When the media device state detector **430** detects a gap in activity of at least a threshold duration (e.g., such as a gap of 4.2 seconds or some other duration, which may be a configuration parameter that can be provided to the media device state detector **430**), the media device state detector **430** sends a probe message on the DDC bus of the HDMI port **405** of the media device **110**. If the media device **110** is switched on (is in the on operational state) the I2C electrical subsystem of the media device **110**, which is implementing the DDC bus of the HDMI port **405**, will acknowledge the probe message regardless of whether any upper layer DDC software is implemented by the media device **110**, or whether any such protocol software, if implemented, has recognized the command represented by the probe message. However, if the media device **110** is switched off (is in the off operational state), the probe message will not be acknowledged by the I2C electrical subsystem of the media device **110**.

In some examples, the probe message is any message, such as an I2C data frame, sent to a particular address of the I2C electrical subsystem implementing the DDC bus of the HDMI port **405**. In some examples, the contents of the probe message are immaterial. For example, the probe message can be an I2C data frame (with any data content) sent to address 0x74 (or some other address, which may be a

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configuration parameter that can be provided to the media device state detector **430** on the DDC bus of the HDMI port **405**. In some such examples, if the media device **110** is in the on operational state, the I2C electrical subsystem of the media device **110** will acknowledge the probe message sent to address 0x74 (or some other address) by pulling down the voltage of an acknowledgment bit that follows the address (and an additional read-write bit) sent on the DDC bus of the HDMI port **405**. If the media device **110** is in the off operational state, the I2C electrical subsystem of the media device **110** will not pull down the voltage of the acknowledgment bit that follows the address (and the additional read-write bit) on the DDC bus of the HDMI port **405** and, thus, will not acknowledge the message sent to address 0x74 (or some other address).

Examples of the acknowledgments monitored by the media device state detector **430** are illustrated in FIGS. 5-6. FIG. 5 illustrates an example of signals monitored by the media device state detector **430** on the DDC bus of the HDMI port **405** of the media device **110**. The signals include an example clock voltage signal **505** sent on the clock pin of the DDC bus of the HDMI port **405** and an example data signal **510** sent on the data pin of the DDC bus of the HDMI port **405**. In the illustrated example of FIG. 5, the clock voltage signal **505** and the example data signal **510** correspond to an example probe message sent by the media device state detector **430** on the DDC bus of the HDMI port **405**. In the illustrated example, the probe message is an I2C data frame that includes a 7-bit address and an additional read-write bit, with one bit signaled at each pulse of the clock signal **505**. In the illustrated example of FIG. 5, the media device **110** is turned on and acknowledges the probe message by pulling down the voltage of an acknowledgment bit of the data signal **510**, which follows the 7-bit address and the additional read-write bit, and which is represented by the circle **515** in the FIG. 5.

FIG. 6 illustrates another example of signals monitored on the DDC bus of the HDMI port **405** by the media device state detector **430**. The signals include an example clock voltage signal **605** sent on the clock pin of the DDC bus of the HDMI port **405** and an example data signal **610** sent on the data pin of the DDC bus of the HDMI port **405**. In the illustrated example of FIG. 6, the clock voltage signal **605** and the example data signal **610** correspond to an example probe message sent by the media device state detector **430** on the DDC bus of the HDMI port **405**. In the illustrated example, the probe message is an I2C data frame that includes a 7-bit address and an additional read-write bit, with one bit signaled at each pulse of the clock signal **605**. In the illustrated example of FIG. 6, the media device **110** is turned off and does not pull down the voltage of an acknowledgment bit of the data signal **610**, which follows the 7-bit address and the additional read-write bit, and which is represented by the circle **615** in the FIG. 6. Thus, in the illustrated example, the media device **110** does not acknowledge the probe message because the media device **110** turned off.

In the illustrated example of FIG. 4, the media device state detector **430** also implements a communication collision avoidance algorithm that avoids collisions on the DDC bus with other connected devices, such as the STB **415**. An example of a communication collision avoidance algorithm implemented by the media device state detector **430** is described in connection with an example implementation of the media device state detector **430** illustrated in FIG. 7.

In some examples, the media device state detector **430** may or may not be a physical part of (e.g., implemented

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by/in) the example meter **114**, as shown in the example of FIG. 4. When the detector **430** is not a physical part of the example meter **114**, the detector **430** may be implemented as a stand-alone device. In some examples, such a stand-alone detector **430** includes HDMI ports to enable the detector **430** to be connected as a pass-through HDMI type of device between the media source (e.g., the STB **415**) and monitored device (e.g., the media device **110**) in a manner similar to that shown in FIG. 4, which enables the detector **430** to intercept the HDMI connection. In some such examples, the detector **430** is also connected to the example meter **114** by a wired connection or wireless connection, such as Bluetooth® (BT), BT low energy (BLE), Wi-Fi, etc., to send the detected operational state of the monitored media device **110** to the meter **114**.

An example implementation of the media device state detector **430** of FIG. 4 is illustrated in FIG. 7. The media device state detector **430** of FIG. 7 may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by processor circuitry such as a central processing unit executing instructions. Additionally or alternatively, the media device state detector **430** of FIG. 7 may be instantiated (e.g., creating an instance of, bring into being for any length of time, materialize, implement, etc.) by an ASIC or an FPGA structured to perform operations corresponding to the instructions. It should be understood that some or all of the circuitry of FIG. 7 may, thus, be instantiated at the same or different times. Some or all of the circuitry may be instantiated, for example, in one or more threads executing concurrently on hardware and/or in series on hardware. Moreover, in some examples, some or all of the circuitry of FIG. 7 may be implemented by one or more virtual machines and/or containers executing on the microprocessor.

The example media device state detector **430** of FIG. 7 includes example bus interface circuitry **705**, example activity detection circuitry **710**, example probe circuitry **715**, example state detection circuitry **720**, and example state output circuitry **725**. In the illustrated example, the bus interface circuitry **705** is structured to interface (e.g., communicate, couple, etc.) with a first of multiple busses of an HDMI port, such as the DDC bus of the HDMI port **405** of the media device **110**. For example, the bus interface circuitry **705** may include one or more of a connector, pins, leads, etc., to electrically couple with the pins of the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**).

In the illustrated example, the activity detection circuitry **710** is to monitor for activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**). For example, as described above in connection with FIGS. 5-6, the activity detection circuitry **710** can monitor for activity (e.g., voltage changes) on the data and/or clock signals of the data and/or clock pins of the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**).

In the illustrated example, the probe circuitry **715** is to inject a probe message with a first address on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**). The probe circuitry **715** is also to detect whether a response to the probe message is received on the monitored bus (e.g., the DDC bus) of the HDMI port **405** of the media device **110**. As described above, the probe message is used to detect the operational state (e.g., on or off) of the monitored media device (e.g., the media device **110**). As also described above in connection with FIGS. 5-6, the probe message sent by the probe

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circuitry **715** can be an I2C data frame with a particular address (such as a 7-bit address with a value of 0x74 or some other address) that is sent on the DDC bus of the HDMI port. As further described above in connection with FIGS. 5-6, the response to the probe message can correspond to an acknowledgment of the probe message that is provided by an I2C electrical subsystem implementing the DDC bus of the HDMI port **405** of the media device **110**. For example, in such examples, the probe circuitry **715** can determine the acknowledgment of the probe message is received when a voltage corresponding to an acknowledgment bit that follows the address of the probe message (and an additional read-write bit) is pulled down on the DDC bus of the HDMI port. In such examples, the probe circuitry **715** can determine the acknowledgment of the probe message has not been received when the voltage corresponding to the acknowledgment bit that follows the address of the probe message (and the additional read-write bit) is not pulled down on the DDC bus of the HDMI port.

In the illustrated example, the state detection circuitry **720** is to detect the operational state of the monitored media device (e.g., the media device **110**) based on whether a response (e.g., an acknowledgment, as described above) to the probe message is received on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**). For example, the state detection circuitry **720** can detect the operational state of the media device to be on when the response to the message is received on the first bus. In such examples, the state detection circuitry **720** can detect the operational state of the media device to be off when the response to the message is not received on the first bus.

In the illustrated example, the state output circuitry **725** is to output the detected operational state of the media device (e.g., the media device **110**) to a meter, such as the meter **114**. For example, the state output circuitry **725** can implement a wired connection or wireless connection, such as BT, BLE, Wi-Fi, etc., to send the detected operational state of the monitored media device (e.g., the media device **110**) to a meter (e.g., the meter **114**).

In the illustrated example of FIG. 7, the activity detection circuitry **710** and the probe circuitry **715** also implement a collision avoidance algorithm, as follows, to avoid collisions on the DDC bus with other connected devices (e.g., such as the STB **415**). In some examples, the activity detection circuitry **710** monitors for activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**), as described above. Then, in response to no activity being detected by the activity detection circuitry **710** on the monitored bus (e.g., the DDC bus of the HDMI port **405** of the media device **110**) for at least a threshold duration, the activity detection circuitry **710** triggers the probe circuitry **715** to inject the probe message described above on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**). In some examples, the threshold duration is a first threshold duration, and in response to no activity being detected by the activity detection circuitry **710** on the monitored bus (e.g., the DDC bus of the HDMI port **405** of the media device **110**) for at least the first threshold duration after the activity detection circuitry **710** (or, more generally, the media device state detector **430**) is initialized or a prior probe message was sent by the probe circuitry **715**, the activity detection circuitry **710** triggers the probe circuitry **715** to inject the probe message described above on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405** of the media device **110**). However, after

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activity is detected by the activity detection circuitry **710** on the monitored bus (e.g., the DDC bus of the HDMI port **405** of the media device **110**), the probe circuitry **715** is triggered to inject the probe message described above on the monitored bus (e.g., the DDC bus of the HDMI port **405** of the media device **110**) in response to no subsequent activity being detected by the activity detection circuitry **710** on the monitored bus (e.g., the DDC bus of the HDMI port **405** of the media device **110**) for at least a second threshold duration. In some examples, the second threshold duration is shorter than the first threshold duration. For example, the first duration may be 4.2 seconds or some other value, whereas the second duration may be 0.2 seconds, or some other value.

In some examples, the media device state detector **430** includes means for monitoring activity of a first bus of an HDMI port of a media device. For example, the means for monitoring may be implemented by the activity detection circuitry **710**. In some examples, the activity detection circuitry **710** may be implemented by machine executable instructions such as that implemented by one or more blocks of FIG. 8 executed by processor circuitry, which may be implemented by the example processor circuitry **912** of FIG. 9, the example processor circuitry **1000** of FIG. 10, and/or the example Field Programmable Gate Array (FPGA) circuitry **1100** of FIG. 11. In other examples, the activity detection circuitry **710** is implemented by other hardware logic circuitry, hardware implemented state machines, and/or any other combination of hardware, software, and/or firmware. For example, the activity detection circuitry **710** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an Application Specific Integrated Circuit (ASIC), a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

In some examples, the media device state detector **430** includes means for inject a message with a first address on a first bus of an HDMI port of a media device. For example, the means for injecting the message may be implemented by the probe circuitry **715**. In some examples, the probe circuitry **715** may be implemented by machine executable instructions such as that implemented by one or more blocks of FIG. 8 executed by processor circuitry, which may be implemented by the example processor circuitry **912** of FIG. 9, the example processor circuitry **1000** of FIG. 10, and/or the example Field Programmable Gate Array (FPGA) circuitry **1100** of FIG. 11. In other examples, the probe circuitry **715** is implemented by other hardware logic circuitry, hardware implemented state machines, and/or any other combination of hardware, software, and/or firmware. For example, the probe circuitry **715** may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an Application Specific Integrated Circuit (ASIC), a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

In some examples, the media device state detector **430** includes means for detecting an operational state of a media device. For example, the means for detecting the operational state may be implemented by the state detection circuitry **720**. In some examples, the state detection circuitry **720** may be implemented by machine executable instructions such as

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that implemented by one or more blocks of FIG. 8 executed by processor circuitry, which may be implemented by the example processor circuitry 912 of FIG. 9, the example processor circuitry 1000 of FIG. 10, and/or the example Field Programmable Gate Array (FPGA) circuitry 1100 of FIG. 11. In other examples, the state detection circuitry 720 is implemented by other hardware logic circuitry, hardware implemented state machines, and/or any other combination of hardware, software, and/or firmware. For example, the state detection circuitry 720 may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an Application Specific Integrated Circuit (ASIC), a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

In some examples, the media device state detector 430 includes means for outputting a detected operational state of a media device. For example, the means for outputting the detected operational state may be implemented by the state output circuitry 725. In some examples, the state output circuitry 725 may be implemented by machine executable instructions such as that implemented by one or more blocks of FIG. 8 executed by processor circuitry, which may be implemented by the example processor circuitry 912 of FIG. 9, the example processor circuitry 1000 of FIG. 10, and/or the example Field Programmable Gate Array (FPGA) circuitry 1100 of FIG. 11. In other examples, the state output circuitry 725 is implemented by other hardware logic circuitry, hardware implemented state machines, and/or any other combination of hardware, software, and/or firmware. For example, the state output circuitry 725 may be implemented by at least one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an Application Specific Integrated Circuit (ASIC), a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

While an example manner of implementing the example meter 114 is illustrated in FIGS. 1-7, one or more of the elements, processes, and/or devices illustrated in FIGS. 1-7 may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, the example media device state detector 430, the example bus interface circuitry 705, the example activity detection circuitry 710, the example probe circuitry 715, the example state detection circuitry 720, the example state output circuitry 725 and/or, more generally, the example meter 114, may be implemented by hardware alone or by hardware in combination with software and/or firmware. Thus, for example, any of the example media device state detector 430, the example bus interface circuitry 705, the example activity detection circuitry 710, the example probe circuitry 715, the example state detection circuitry 720, the example state output circuitry 725 and/or, more generally, the example meter 114 could be implemented by processor circuitry, analog circuit(s), digital circuit(s), logic circuit(s), programmable processor(s), programmable microcontroller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)), and/or field programmable logic device(s) (FPLD(s)) such as Field Programmable Gate Arrays (FPGAs). Further still, the example meter 114 may include one or more elements, processes and/or devices in addition to, or instead of, those illustrated

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in FIGS. 1-7, and/or may include more than one of any or all of the illustrated elements, processes and devices. As used herein, the phrase "in communication," including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

A flowchart representative of example hardware logic circuitry, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing the example meter 114, the example media device state detector 430, the example bus interface circuitry 705, the example activity detection circuitry 710, the example probe circuitry 715, the example state detection circuitry 720, and/or the example state output circuitry 725 is shown in FIG. 8. In these examples, the machine readable instructions may be one or more executable programs or portion(s) of an executable program for execution by a computer processor and/or processor circuitry, such as the processor circuitry 912 shown in the example processor platform 900 discussed below in connection with FIG. 9 and/or the example processor circuitry discussed below in connection with FIGS. 10 and/or 11. The one or more programs, or portion(s) thereof, may be embodied in software stored on one or more non-transitory computer readable storage media such as compact disk (CD), a floppy disk, a hard disk drive (HDD), a solid-state drive (SSD), a digital versatile disk (DVD), a Blu-ray disk, a volatile memory (e.g., Random Access Memory (RAM) of any type, etc.), or a non-volatile memory (e.g., electrically erasable programmable read-only memory (EEPROM), FLASH memory, an HDD, an SSD, etc.) associated with processor circuitry located in one or more hardware devices, but the entire program or programs and/or parts thereof could alternatively be executed by one or more hardware devices other than the processor circuitry and/or embodied in firmware or dedicated hardware. The machine readable instructions may be distributed across multiple hardware devices and/or executed by two or more hardware devices (e.g., a server and a client hardware device). For example, the client hardware device may be implemented by an endpoint client hardware device (e.g., a hardware device associated with a user) or an intermediate client hardware device (e.g., a radio access network (RAN) gateway) that may facilitate communication between a server and an endpoint client hardware device). Similarly, the non-transitory computer readable storage media may include one or more mediums located in one or more hardware devices. Further, although the example program(s) is(are) described with reference to the flowcharts illustrated in FIG. 8, many other methods of implementing the example meter 114, the example media device state detector 430, the example bus interface circuitry 705, the example activity detection circuitry 710, the example probe circuitry 715, the example state detection circuitry 720, and/or the example state output circuitry 725 may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, combined and/or subdivided into multiple blocks. Additionally or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., processor circuitry, discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding

operation without executing software or firmware. The processor circuitry may be distributed in different network locations and/or local to one or more hardware devices (e.g., a single-core processor (e.g., a single core central processor unit (CPU)), a multi-core processor (e.g., a multi-core CPU), etc.) in a single machine, multiple processors distributed across multiple servers of a server rack, multiple processors distributed across one or more server racks, a CPU and/or a FPGA located in the same package (e.g., the same integrated circuit (IC) package or in two or more separate housings, etc.).

The machine readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine readable instructions as described herein may be stored as data or a data structure (e.g., as portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine readable instructions may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The machine readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc., in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and/or stored on separate computing devices, wherein the parts when decrypted, decompressed, and/or combined form a set of machine executable instructions that implement one or more operations that may together form a program such as that described herein.

In another example, the machine readable instructions may be stored in a state in which they may be read by processor circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), a software development kit (SDK), an application programming interface (API), etc., in order to execute the machine readable instructions on a particular computing device or other device. In another example, the machine readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine readable media, as used herein, may include machine readable instructions and/or program(s) regardless of the particular format or state of the machine readable instructions and/or program(s) when stored or otherwise at rest or in transit.

The machine readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine readable instructions may be represented using any of the following languages: C, C++, Java, C #, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

As mentioned above, the example operations of FIG. 8 may be implemented using executable instructions (e.g., computer and/or machine readable instructions) stored on one or more non-transitory computer and/or machine readable media such as optical storage devices, magnetic storage devices, an HDD, a flash memory, a read-only memory

(ROM), a CD, a DVD, a cache, a RAM of any type, a register, and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the terms non-transitory computer readable medium and non-transitory computer readable storage medium are expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media. Also, as used herein, the terms “computer readable” and “machine readable” are considered equivalent unless indicated otherwise.

“Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.

As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

FIG. 8 is a flowchart representative of example machine readable instructions and/or example operations 800 that may be executed and/or instantiated by processor circuitry to implement the media device state detector 430 or, more generally, the meter 114 described above. With reference to the preceding figures and associated written descriptions, the

machine readable instructions and/or operations **800** of FIG. **8** begin at block **805** at which the activity detection circuitry **710** of the media device state detector **430** resets an activity timer used by the media device state detector **430** to monitor for periods of inactivity on a monitored bus (e.g., the DDC bus) of an HDMI port (e.g., the HDMI port **405**) of a media device (e.g., the media device **110**). The activity timer can be any type of timer, clock, etc., and/or other circuitry implemented and/or accessible by the activity detection circuitry **710**. In some examples, the instructions and/or operations **800** reach block **805** after the activity detection circuitry **710**, the media device state detector **430** and/or the meter **114** is initialized, reset, powered-on, etc. Additionally or alternatively, in some examples, the instructions and/or operations **800** reach block **805** after a prior probe message has been injected on the monitored bus (e.g., the DDC bus) of an HDMI port (e.g., the HDMI port **405**) of a media device (e.g., the media device **110**) by the probe circuitry **715** of the media device state detector **430**, and/or after an operation state of the media device (e.g., the media device **110**) has been detected based on the prior probe message by the state detection circuitry **720** of the media device state detector **430**.

At block **810**, the activity detection circuitry **710** monitors for activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**. For example, and as described above, the activity detection circuitry **710** can monitor for activity (e.g., voltage changes) on the data and clock signals of the data and clock pins of the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**.

At block **815**, the activity detection circuitry **710** determines whether activity has been detected on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**. If no activity is detected by the activity detection circuitry **710** (corresponding to the “NO” branch out of block **815**), then at block **820** the activity detection circuitry **710** evaluates the activity timer to determine whether a first threshold inactivity duration has expired. For example, the first threshold inactivity duration can be set to 4.2 seconds or some other threshold duration, which may be a variable parameter that is configurable based on user input, a fixed, preset value, a compilation parameter, etc.

If the activity detection circuitry **710** determines the first threshold inactivity duration has not expired (corresponding to the “NO” branch out of block **820**), control returns to block **815** at which the activity detection circuitry **710** continues to monitor for activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**. However, if the activity detection circuitry **710** determines the first threshold inactivity duration has expired (corresponding to the “YES” branch out of block **820**), then at block **825** the activity detection circuitry **710** triggers the probe circuitry **715** of the media device state detector **430** to inject a probe message onto the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**), as described above.

At block **830**, the probe circuitry **715** determines whether a response to the injected probe message is received on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**), as described above. If the probe circuitry **715** determine a response to the probe message has been received (corresponding to the “YES” branch out of block **830**), then at block **835** the state detection circuitry **720** of the media device state detector **430** sets the detected operational state of the monitored

media device (e.g., the media device **110**) to on (or an on state, an on operational state, an active state, etc.). However, if the probe circuitry **715** determine a response to the probe message has not been received (corresponding to the “NO” branch out of block **830**), then at block **840** the state detection circuitry **720** sets the detected operational state of the monitored media device (e.g., the media device **110**) to off (or an off state, an off operational state, an inactive state, etc.). In some examples, at block **835** and **840**, the state output circuitry **725** of the media device state detector **430** outputs the detected operational state of the monitored media device (e.g., the media device **110**) to a meter (e.g., the meter **114**).

After the operational state of the monitored media device (e.g., the media device **110**) is detected by the state detection circuitry **720** (blocks **835** or **840**), at block **845** the state detection circuitry **720** determines whether monitoring is to end. If monitoring is to end (corresponding to the “YES” branch out of block **845**), then the machine readable instructions and/or operations **800** end. However, if monitoring is to continue (corresponding to the “NO” branch out of block **845**), control returns to block **805** and blocks subsequent thereto.

Returning to block **815**, if activity is detected by the activity detection circuitry **710** on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110** (corresponding to the “YES” branch out of block **815**), then at block **850** the activity detection circuitry **710** resets the activity timer. At block **855**, the activity detection circuitry **710** continues to monitor for subsequent activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**. If subsequent activity is detected by the activity detection circuitry **710** (corresponding to the “YES” branch out of block **855**), control returns to block **850** at which the activity detection circuitry **710** resets the activity timer and continues to monitor for subsequent activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**). However, if no subsequent activity is detected by the activity detection circuitry **710** (corresponding to the “NO” branch out of block **855**), then at block **860** the activity detection circuitry **710** evaluates the activity timer to determine whether a second threshold inactivity duration has expired. For example, the second threshold inactivity duration can be set to 0.2 seconds or some other threshold duration, which may be a variable parameter that is configurable based on user input, a fixed, preset value, a compilation parameter, etc.

If the activity detection circuitry **710** determines the second threshold inactivity duration has not expired (corresponding to the “NO” branch out of block **860**), control returns to block **855** at which the activity detection circuitry **710** continues to monitor for subsequent activity on the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**). However, if the activity detection circuitry **710** determines the second threshold inactivity duration has expired (corresponding to the “YES” branch out of block **860**), then control proceeds to block **825** at which the activity detection circuitry **710** triggers the probe circuitry **715** of the media device state detector **430** to inject a probe message onto the monitored bus (e.g., the DDC bus) of the HDMI port (e.g., the HDMI port **405**) of the media device **110**), as described above. The machine readable instructions and/or operations **800** then proceed as described above.

FIG. **9** is a block diagram of an example processor platform **900** structured to execute and/or instantiate the

machine readable instructions and/or the operations of FIG. 8 to implement the example meter 114 and/or the example media device state detector 430 of FIGS. 1-7. The processor platform 900 can be, for example, a server, a personal computer, a workstation, a self-learning machine (e.g., a neural network), a mobile device (e.g., a cell phone, a smart phone, a tablet such as an iPad™), a personal digital assistant (PDA), an Internet appliance, a DVD player, a CD player, a digital video recorder, a Blu-ray player, a gaming console, a personal video recorder, a set top box, a headset (e.g., an augmented reality (AR) headset, a virtual reality (VR) headset, etc.) or other wearable device, or any other type of computing device.

The processor platform 900 of the illustrated example includes a processor 912. The processor 912 of the illustrated example is hardware. For example, the processor 912 can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUS, DSPs, or controllers from any desired family or manufacturer. The hardware processor 912 may be a semiconductor based (e.g., silicon based) device. In this example, the processor 912 implements the example bus interface circuitry 705, the example activity detection circuitry 710, the example probe circuitry 715, the example state detection circuitry 720, and/or the example state output circuitry 725 of the media device state detector 430.

The processor 912 of the illustrated example includes a local memory 913 (e.g., a cache, registers, etc.). The processor circuitry 912 of the illustrated example is in communication with a main memory including a volatile memory 914 and a non-volatile memory 916 via a link 918. The link 918 may be implemented by a bus, one or more point-to-point connections, etc., or a combination thereof. The volatile memory 914 may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of RAM device. The non-volatile memory 916 may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory 914, 916 of the illustrated example is controlled by a memory controller 917.

The processor platform 900 of the illustrated example also includes interface circuitry 920. The interface circuitry 920 may be implemented by hardware in accordance with any type of interface standard, such as an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a Peripheral Component Interconnect (PCI) interface, and/or a Peripheral Component Interconnect Express (PCIe) interface.

In the illustrated example, one or more input devices 922 are connected to the interface circuitry 920. The input device(s) 922 permit(s) a user to enter data and/or commands into the processor circuitry 912. The input device(s) can be implemented by, for example, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, a trackbar (such as an isopoint device), a voice recognition system and/or any other human-machine interface. Also, many systems, such as the processor platform 900, can allow the user to control the computer system and provide data to the computer using physical gestures, such as, but not limited to, hand or body movements, facial expressions, and face recognition.

One or more output devices 924 are also connected to the interface circuitry 920 of the illustrated example. The output device(s) 924 can be implemented, for example, by display

devices (e.g., a light emitting diode (LED), an organic light emitting diode (OLED), a liquid crystal display (LCD), a cathode ray tube (CRT) display, an in-place switching (IPS) display, a touchscreen, etc.), a tactile output device, a printer and/or speakers(s). The interface circuitry 920 of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or graphics processor circuitry such as a GPU.

The interface circuitry 920 of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) by a network 926. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, an optical connection, etc.

In some examples, the interface circuit 920 is configured to implement the example HDMI ports 340/or 410 of the meter 114.

The processor platform 900 of the illustrated example also includes one or more mass storage devices 928 to store software and/or data. Examples of such mass storage devices 928 include magnetic storage devices, optical storage devices, floppy disk drives, HDDs, CDs, Blu-ray disk drives, redundant array of independent disks (RAID) systems, solid state storage devices such as flash memory devices and/or SSDs, and DVD drives.

The machine executable instructions 932 which may be implemented by the machine readable instructions of FIG. 8 may be stored in the mass storage device 928, in the volatile memory 914, in the non-volatile memory 916, in the local memory 913 and/or on a removable non-transitory computer readable storage medium, such as a CD or DVD 936.

FIG. 10 is a block diagram of an example implementation of the processor circuitry 912 of FIG. 9. In this example, the processor circuitry 912 of FIG. 9 is implemented by a microprocessor 1000. The general purpose microprocessor circuitry 1000 executes some or all of the machine readable instructions of the flowchart of FIG. 8 to effectively instantiate the circuitry of FIG. 7 as logic circuits to perform the operations corresponding to those machine readable instructions. In some such examples, the circuitry of FIG. 7 is instantiated by the hardware circuits of the microprocessor 1000 in combination with the instructions. For example, the microprocessor 1000 may implement multi-core hardware circuitry such as a CPU, a DSP, a GPU, an XPU, etc. Although it may include any number of example cores 1002 (e.g., 1 core), the microprocessor 1000 of this example is a multi-core semiconductor device including N cores. The cores 1002 of the microprocessor 1000 may operate independently or may cooperate to execute machine readable instructions. For example, machine code corresponding to a firmware program, an embedded software program, or a software program may be executed by one of the cores 1002 or may be executed by multiple ones of the cores 1002 at the same or different times. In some examples, the machine code corresponding to the firmware program, the embedded software program, or the software program is split into threads and executed in parallel by two or more of the cores 1002. The software program may correspond to a portion or all of the machine readable instructions and/or operations represented by the flowchart of FIG. 8.

The cores 1002 may communicate by a first example bus 1004. In some examples, the first bus 1004 may implement

a communication bus to effectuate communication associated with one(s) of the cores **1002**. For example, the first bus **1004** may implement at least one of an Inter-Integrated Circuit (I2C) bus, a Serial Peripheral Interface (SPI) bus, a PCI bus, or a PCIe bus. Additionally or alternatively, the first bus **1004** may implement any other type of computing or electrical bus. The cores **1002** may obtain data, instructions, and/or signals from one or more external devices by example interface circuitry **1006**. The cores **1002** may output data, instructions, and/or signals to the one or more external devices by the interface circuitry **1006**. Although the cores **1002** of this example include example local memory **1020** (e.g., Level 1 (L1) cache that may be split into an L1 data cache and an L1 instruction cache), the microprocessor **1000** also includes example shared memory **1010** that may be shared by the cores (e.g., Level 2 (L2) cache) for high-speed access to data and/or instructions. Data and/or instructions may be transferred (e.g., shared) by writing to and/or reading from the shared memory **1010**. The local memory **1020** of each of the cores **1002** and the shared memory **1010** may be part of a hierarchy of storage devices including multiple levels of cache memory and the main memory (e.g., the main memory **914**, **916** of FIG. 9). Typically, higher levels of memory in the hierarchy exhibit lower access time and have smaller storage capacity than lower levels of memory. Changes in the various levels of the cache hierarchy are managed (e.g., coordinated) by a cache coherency policy.

Each core **1002** may be referred to as a CPU, DSP, GPU, etc., or any other type of hardware circuitry. Each core **1002** includes control unit circuitry **1014**, arithmetic and logic (AL) circuitry (sometimes referred to as an ALU) **1016**, a plurality of registers **1018**, the L1 cache **1020**, and a second example bus **1022**. Other structures may be present. For example, each core **1002** may include vector unit circuitry, single instruction multiple data (SIMD) unit circuitry, load/store unit (LSU) circuitry, branch/jump unit circuitry, floating-point unit (FPU) circuitry, etc. The control unit circuitry **1014** includes semiconductor-based circuits structured to control (e.g., coordinate) data movement within the corresponding core **1002**. The AL circuitry **1016** includes semiconductor-based circuits structured to perform one or more mathematic and/or logic operations on the data within the corresponding core **1002**. The AL circuitry **1016** of some examples performs integer based operations. In other examples, the AL circuitry **1016** also performs floating point operations. In yet other examples, the AL circuitry **1016** may include first AL circuitry that performs integer based operations and second AL circuitry that performs floating point operations. In some examples, the AL circuitry **1016** may be referred to as an Arithmetic Logic Unit (ALU). The registers **1018** are semiconductor-based structures to store data and/or instructions such as results of one or more of the operations performed by the AL circuitry **1016** of the corresponding core **1002**. For example, the registers **1018** may include vector register(s), SIMD register(s), general purpose register(s), flag register(s), segment register(s), machine specific register(s), instruction pointer register(s), control register(s), debug register(s), memory management register(s), machine check register(s), etc. The registers **1018** may be arranged in a bank as shown in FIG. 10. Alternatively, the registers **1018** may be organized in any other arrangement, format, or structure including distributed throughout the core **1002** to shorten access time. The second bus **1020** may implement at least one of an I2C bus, a SPI bus, a PCI bus, or a PCIe bus.

Each core **1002** and/or, more generally, the microprocessor **1000** may include additional and/or alternate structures to those shown and described above. For example, one or more clock circuits, one or more power supplies, one or more power gates, one or more cache home agents (CHAs), one or more converged/common mesh stops (CMSs), one or more shifters (e.g., barrel shifter(s)) and/or other circuitry may be present. The microprocessor **1000** is a semiconductor device fabricated to include many transistors interconnected to implement the structures described above in one or more integrated circuits (ICs) contained in one or more packages. The processor circuitry may include and/or cooperate with one or more accelerators. In some examples, accelerators are implemented by logic circuitry to perform certain tasks more quickly and/or efficiently than can be done by a general purpose processor. Examples of accelerators include ASICs and FPGAs such as those discussed herein. A GPU or other programmable device can also be an accelerator. Accelerators may be on-board the processor circuitry, in the same chip package as the processor circuitry and/or in one or more separate packages from the processor circuitry.

FIG. 11 is a block diagram of another example implementation of the processor circuitry **912** of FIG. 9. In this example, the processor circuitry **912** is implemented by FPGA circuitry **1100**. The FPGA circuitry **1100** can be used, for example, to perform operations that could otherwise be performed by the example microprocessor **1000** of FIG. 10 executing corresponding machine readable instructions. However, once configured, the FPGA circuitry **1000** instantiates the machine readable instructions in hardware and, thus, can often execute the operations faster than they could be performed by a general purpose microprocessor executing the corresponding software.

More specifically, in contrast to the microprocessor **1000** of FIG. 10 described above (which is a general purpose device that may be programmed to execute some or all of the machine readable instructions represented by the flowchart of FIG. 8 but whose interconnections and logic circuitry are fixed once fabricated), the FPGA circuitry **1100** of the example of FIG. 11 includes interconnections and logic circuitry that may be configured and/or interconnected in different ways after fabrication to instantiate, for example, some or all of the machine readable instructions represented by the flowchart of FIG. 8. In particular, the FPGA **1100** may be thought of as an array of logic gates, interconnections, and switches. The switches can be programmed to change how the logic gates are interconnected by the interconnections, effectively forming one or more dedicated logic circuits (unless and until the FPGA circuitry **1100** is reprogrammed). The configured logic circuits enable the logic gates to cooperate in different ways to perform different operations on data received by input circuitry. Those operations may correspond to some or all of the software represented by the flowchart of FIG. 8. As such, the FPGA circuitry **1100** may be structured to effectively instantiate some or all of the machine readable instructions of the flowchart of FIG. 8 as dedicated logic circuits to perform the operations corresponding to those software instructions in a dedicated manner analogous to an ASIC. Therefore, the FPGA circuitry **1100** may perform the operations corresponding to the some or all of the machine readable instructions of FIG. 8 faster than the general purpose microprocessor can execute the same.

In the example of FIG. 11, the FPGA circuitry **1100** is structured to be programmed (and/or reprogrammed one or more times) by an end user by a hardware description

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language (HDL) such as Verilog. The FPGA circuitry **1100** of FIG. **11** includes example input/output (I/O) circuitry **1102** to obtain and/or output data to/from example configuration circuitry **1104** and/or external hardware (e.g., external hardware circuitry) **1106**. For example, the configuration circuitry **1104** may implement interface circuitry that may obtain machine readable instructions to configure the FPGA circuitry **1100**, or portion(s) thereof. In some such examples, the configuration circuitry **1104** may obtain the machine readable instructions from a user, a machine (e.g., hardware circuitry (e.g., programmed or dedicated circuitry) that may implement an Artificial Intelligence/Machine Learning (AI/ML) model to generate the instructions), etc. In some examples, the external hardware **1106** may implement the microprocessor **1000** of FIG. **10**. The FPGA circuitry **1100** also includes an array of example logic gate circuitry **1108**, a plurality of example configurable interconnections **1110**, and example storage circuitry **1112**. The logic gate circuitry **1108** and interconnections **1110** are configurable to instantiate one or more operations that may correspond to at least some of the machine readable instructions of FIG. **8** and/or other desired operations. The logic gate circuitry **1108** shown in FIG. **11** is fabricated in groups or blocks. Each block includes semiconductor-based electrical structures that may be configured into logic circuits. In some examples, the electrical structures include logic gates (e.g., And gates, Or gates, Nor gates, etc.) that provide basic building blocks for logic circuits. Electrically controllable switches (e.g., transistors) are present within each of the logic gate circuitry **1108** to enable configuration of the electrical structures and/or the logic gates to form circuits to perform desired operations. The logic gate circuitry **1108** may include other electrical structures such as look-up tables (LUTs), registers (e.g., flip-flops or latches), multiplexers, etc.

The interconnections **1110** of the illustrated example are conductive pathways, traces, vias, or the like that may include electrically controllable switches (e.g., transistors) whose state can be changed by programming (e.g., using an HDL instruction language) to activate or deactivate one or more connections between one or more of the logic gate circuitry **1108** to program desired logic circuits.

The storage circuitry **1112** of the illustrated example is structured to store result(s) of the one or more of the operations performed by corresponding logic gates. The storage circuitry **1112** may be implemented by registers or the like. In the illustrated example, the storage circuitry **1112** is distributed amongst the logic gate circuitry **1108** to facilitate access and increase execution speed.

The example FPGA circuitry **1100** of FIG. **11** also includes example Dedicated Operations Circuitry **1114**. In this example, the Dedicated Operations Circuitry **1114** includes special purpose circuitry **1116** that may be invoked to implement commonly used functions to avoid the need to program those functions in the field. Examples of such special purpose circuitry **1116** include memory (e.g., DRAM) controller circuitry, PCIe controller circuitry, clock circuitry, transceiver circuitry, memory, and multiplier-accumulator circuitry. Other types of special purpose circuitry may be present. In some examples, the FPGA circuitry **1100** may also include example general purpose programmable circuitry **1118** such as an example CPU **1120** and/or an example DSP **1122**. Other general purpose programmable circuitry **1118** may additionally or alternatively be present such as a GPU, an XPU, etc., that can be programmed to perform other operations.

Although FIGS. **10** and **11** illustrate two example implementations of the processor circuitry **912** of FIG. **9**, many

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other approaches are contemplated. For example, as mentioned above, modern FPGA circuitry may include an on-board CPU, such as one or more of the example CPU **1120** of FIG. **11**. Therefore, the processor circuitry **912** of FIG. **9** may additionally be implemented by combining the example microprocessor **1000** of FIG. **10** and the example FPGA circuitry **1100** of FIG. **11**. In some such hybrid examples, a first portion of the machine readable instructions represented by the flowchart of FIG. **8** may be executed by one or more of the cores **1002** of FIG. **10**, a second portion of the machine readable instructions represented by the flowchart of FIG. **8** may be executed by the FPGA circuitry **1100** of FIG. **11**, and/or a third portion of the machine readable instructions represented by the flowchart of FIG. **8** may be executed by an ASIC. It should be understood that some or all of the circuitry of FIG. **7** may, thus, be instantiated at the same or different times. Some or all of the circuitry may be instantiated, for example, in one or more threads executing concurrently and/or in series. Moreover, in some examples, some or all of the circuitry of FIG. **7** may be implemented within one or more virtual machines and/or containers executing on the microprocessor.

In some examples, the processor circuitry **912** of FIG. **9** may be in one or more packages. For example, the processor circuitry **1000** of FIG. **10** and/or the FPGA circuitry **1100** of FIG. **11** may be in one or more packages. In some examples, an XPU may be implemented by the processor circuitry **912** of FIG. **9**, which may be in one or more packages. For example, the XPU may include a CPU in one package, a DSP in another package, a GPU in yet another package, and an FPGA in still yet another package.

A block diagram illustrating an example software distribution platform **1205** to distribute software such as the example machine readable instructions **932** of FIG. **9** to hardware devices owned and/or operated by third parties is illustrated in FIG. **12**. The example software distribution platform **1205** may be implemented by any computer server, data facility, cloud service, etc., capable of storing and transmitting software to other computing devices. The third parties may be customers of the entity owning and/or operating the software distribution platform **1205**. For example, the entity that owns and/or operates the software distribution platform **1205** may be a developer, a seller, and/or a licensor of software such as the example machine readable instructions **932** of FIG. **9**. The third parties may be consumers, users, retailers, OEMs, etc., who purchase and/or license the software for use and/or re-sale and/or sub-licensing. In the illustrated example, the software distribution platform **1205** includes one or more servers and one or more storage devices. The storage devices store the machine readable instructions **932**, which may correspond to the example machine readable instructions **800** of FIG. **8**, as described above. The one or more servers of the example software distribution platform **1205** are in communication with a network **1210**, which may correspond to any one or more of the Internet and/or any of the example networks **180** and/or **926** described above. In some examples, the one or more servers are responsive to requests to transmit the software to a requesting party as part of a commercial transaction. Payment for the delivery, sale, and/or license of the software may be handled by the one or more servers of the software distribution platform and/or by a third party payment entity. The servers enable purchasers and/or licensors to download the machine readable instructions **932** from the software distribution platform **1205**. For example, the software, which may correspond to the example machine readable instructions **800** of FIG. **8**, may be downloaded to

the example processor platform **900**, which is to execute the machine readable instructions **932** to implement the example meter **114** and/or the example media device state detector **430**. In some example, one or more servers of the software distribution platform **1205** periodically offer, transmit, and/or force updates to the software (e.g., the example machine readable instructions **932** of FIG. **9**) to ensure improvements, patches, updates, etc., are distributed and applied to the software at the end user devices.

From the foregoing, it will be appreciated that example systems, methods, apparatus, and articles of manufacture have been disclosed that detect an operational state (such as an on/off state) of a media device. Disclosed systems, methods, apparatus, and articles of manufacture improve the efficiency of using a computing device by utilizing an HDMI port of a monitored media device (e.g., TV) to detect an operational state of the monitored media device without relying on any particular protocol commands being implemented over the HDMI port to detect the operational state of the monitored media device (e.g., TV). Disclosed systems, methods, apparatus, and articles of manufacture are accordingly directed to one or more improvement(s) in the operation of a machine such as a computer or other electronic and/or mechanical device.

Example methods, apparatus, systems, and articles of manufacture to detect an operational state of a media device are disclosed herein. Further examples and combinations thereof include the following.

Example 1 includes an apparatus to detect an operational state of a media device, the apparatus comprising activity detection circuitry to monitor for activity on a first bus of an (HDMI) port of the media device, probe circuitry to inject a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration, and state detection circuitry to detect the operational state of the media device based on whether a response to the message is received on the first bus.

Example 2 includes the apparatus of example 1, wherein the state detection circuitry is to detect the operational state of the media device to be on when the response to the message is received on the first bus, and detect the operational state of the media device to be off when the response to the message is not received on the first bus.

Example 3 includes the apparatus of example 1, wherein the first bus is a display data channel (DDC) bus of the HDMI port.

Example 4 includes the apparatus of example 3, wherein the response corresponds to an acknowledgment of the message that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

Example 5 includes the apparatus of example 4, wherein the probe circuitry is to determine the acknowledgment of the message is received when a voltage corresponding to an acknowledgment bit that follows the first address and a read-write bit is pulled down on the DDC bus of the HDMI port.

Example 6 includes the apparatus of example 1, wherein the threshold duration is a first threshold duration, and after activity is detected by the activity detection circuitry on the first bus of the HDMI port of the media device, the probe circuitry is to inject the message with the first address on the first bus in response to detection of no subsequent activity on the first bus for at least a second threshold duration.

Example 7 includes the apparatus of example 6, wherein the second threshold duration is shorter than the first threshold duration.

Example 8 includes the apparatus of example 1, further including state output circuitry to output the detected operational state of the media device to a meter.

Example 9 includes an apparatus to detect an operational state of a media device, the apparatus comprising at least one memory, instructions in the apparatus, and processor circuitry to execute the instructions to at least monitor for activity on a first bus of an (HDMI) port of the media device, inject a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration, and detect the operational state of the media device based on whether a response to the message is received on the first bus.

Example 10 includes the apparatus of example 9, wherein the processor circuitry is to detect the operational state of the media device to be on when the response to the message is received on the first bus, and detect the operational state of the media device to be off when the response to the message is not received on the first bus.

Example 11 includes the apparatus of example 9, wherein the first bus is a display data channel (DDC) bus of the HDMI port.

Example 12 includes the apparatus of example 11, wherein the response corresponds to an acknowledgment of the message that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

Example 13 includes the apparatus of example 12, wherein the processor circuitry is to determine the acknowledgment of the message is received when a voltage corresponding to an acknowledgment bit that follows the first address and a read-write bit is pulled down on the DDC bus of the HDMI port.

Example 14 includes the apparatus of example 9, wherein the threshold duration is a first threshold duration, and after activity is detected on the first bus of the HDMI port of the media device, the processor circuitry is to inject the message with the first address on the first bus in response to detection of no subsequent activity on the first bus for at least a second threshold duration.

Example 15 includes the apparatus of example 14, wherein the second threshold duration is shorter than the first threshold duration.

Example 16 includes the apparatus of example 9, wherein the processor circuitry is to output the detected operational state of the media device to a meter.

Example 17 includes at least one non-transitory computer readable medium comprising computer readable instructions that, when executed, cause at least one processor to at least monitor for activity on a first bus of an (HDMI) port of a media device, inject a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration, and detect an operational state of the media device based on whether a response to the message is received on the first bus.

Example 18 includes the at least one non-transitory computer readable medium of example 17, wherein the instructions cause the at least one processor to detect the operational state of the media device to be on when the response to the message is received on the first bus, and detect the operational state of the media device to be off when the response to the message is not received on the first bus.

Example 19 includes the at least one non-transitory computer readable medium of example 17, wherein the first bus is a display data channel (DDC) bus of the HDMI port.

Example 20 includes the at least one non-transitory computer readable medium of example 19, wherein the response corresponds to an acknowledgment of the message that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

Example 21 includes the at least one non-transitory computer readable medium of example 20, wherein the instructions cause the at least one processor to determine the acknowledgment of the message is received when a voltage corresponding to an acknowledgment bit that follows the first address and a read-write bit is pulled down on the DDC bus of the HDMI port.

Example 22 includes the at least one non-transitory computer readable medium of example 17, wherein the threshold duration is a first threshold duration, and after activity is detected on the first bus of the HDMI port of the media device, the instructions cause the at least one processor to inject the message with the first address on the first bus in response to detection of no subsequent activity on the first bus for at least a second threshold duration.

Example 23 includes the at least one non-transitory computer readable medium of example 22, wherein the second threshold duration is shorter than the first threshold duration.

Example 24 includes the at least one non-transitory computer readable medium of example 17, wherein instructions cause the at least one processor to output the detected operational state of the media device to a meter.

Example 25 includes a method to detect an operational state of a media device, the method comprising monitoring for activity on a first bus of an (HDMI) port of the media device, injecting a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration, and detecting the operational state of the media device based on whether a response to the message is received on the first bus.

Example 26 includes the method of example 25, wherein the detecting of the operational state of the media device includes detecting the operational state of the media device to be on when the response to the message is received on the first bus, and detecting the operational state of the media device to be off when the response to the message is not received on the first bus.

Example 27 includes the method of example 25, wherein the first bus is a display data channel (DDC) bus of the HDMI port.

Example 28 includes the method of example 27, wherein the response corresponds to an acknowledgment of the message that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

Example 29 includes the method of example 28, further including determining the acknowledgment of the message is received when a voltage corresponding to an acknowledgment bit that follows the first address and a read-write bit is pulled down on the DDC bus of the HDMI port.

Example 30 includes the method of example 25, wherein the threshold duration is a first threshold duration, and further including, in response to activity being detected on the first bus of the HDMI port of the media device, injecting the message with the first address on the first bus in response to detection of no subsequent activity on the first bus for at least a second threshold duration.

Example 31 includes the method of example 30, wherein the second threshold duration is shorter than the first threshold duration.

Example 32 includes the method of example 25, further including outputting the detected operational state of the media device to a meter.

Example 33 includes a system to detect an operational state of a media device, the system comprising means for monitoring for activity on a first bus of an (HDMI) port of the media device, means for injecting a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration, and means for detecting the operational state of the media device based on whether a response to the message is received on the first bus.

Example 34 includes the system of example 33, wherein the means for detecting is to detect the operational state of the media device to be on when the response to the message is received on the first bus, and detect the operational state of the media device to be off when the response to the message is not received on the first bus.

Example 35 includes the system of example 33, wherein the first bus is a display data channel (DDC) bus of the HDMI port.

Example 36 includes the system of example 35, wherein the response corresponds to an acknowledgment of the message that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

Example 37 includes the system of example 36, wherein the means for injecting the message is to determine the acknowledgment of the message is received when a voltage corresponding to an acknowledgment bit that follows the first address and a read-write bit is pulled down on the DDC bus of the HDMI port.

Example 38 includes the system of example 33, wherein the threshold duration is a first threshold duration, and in response to activity being detected on the first bus of the HDMI port of the media device, the means for injecting the message is to inject the message with the first address on the first bus in response to detection of no subsequent activity on the first bus for at least a second threshold duration.

Example 39 includes the system of example 38, wherein the second threshold duration is shorter than the first threshold duration.

Example 40 includes the system of example 33, further including means for outputting the detected operational state of the media device to a meter.

Example 41 includes an apparatus to detect an operational state of a media device, the apparatus comprising interface circuitry to electrically couple with a first bus of an (HDMI) port of the media device, and processor circuitry including one or more of at least one of a central processing unit, a graphic processing unit, or a digital signal processor, the at least one of the central processing unit, the graphic processing unit, or the digital signal processor having control circuitry to control data movement within the processor circuitry, arithmetic and logic circuitry to perform one or more first operations corresponding to instructions, and one or more registers to store a result of the one or more first operations, the instructions in the apparatus, a Field Programmable Gate Array (FPGA), the FPGA including logic gate circuitry, a plurality of configurable interconnections, and storage circuitry, the logic gate circuitry and interconnections to perform one or more second operations, the storage circuitry to store a result of the one or more second operations, or Application Specific Integrate Circuitry (ASIC) including logic gate circuitry to perform one or more third operations, the processor circuitry to perform at least one of the first operations, the second operations, or the third

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operations to instantiate activity detection circuitry to monitor for activity on a first bus of an (HDMI) port of the media device, probe circuitry to inject a message with a first address on the first bus in response to detection of no activity on the first bus for at least a threshold duration, and state detection circuitry to detect the operational state of the media device based on whether a response to the message is received on the first bus.

Example 42 includes the apparatus of example 41, wherein the state detection circuitry is to detect the operational state of the media device to be on when the response to the message is received on the first bus, and detect the operational state of the media device to be off when the response to the message is not received on the first bus.

Example 43 includes the apparatus of example 41, wherein the first bus is a display data channel (DDC) bus of the HDMI port.

Example 44 includes the apparatus of example 43, wherein the response corresponds to an acknowledgment of the message that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

Example 45 includes the apparatus of example 44, wherein the probe circuitry is to determine the acknowledgment of the message is received when a voltage corresponding to an acknowledgment bit that follows the first address and a read-write bit is pulled down on the DDC bus of the HDMI port.

Example 46 includes the apparatus of example 41, wherein the threshold duration is a first threshold duration, and after activity is detected by the activity detection circuitry on the first bus of the HDMI port of the media device, the probe circuitry is to inject the message with the first address on the first bus in response to detection of no subsequent activity on the first bus for at least a second threshold duration.

Example 47 includes the apparatus of example 46, wherein the second threshold duration is shorter than the first threshold duration.

Example 48 includes the apparatus of example 41, further including state output circuitry to output the detected operational state of the media device to a meter.

The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, methods, apparatus and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all methods, apparatus and articles of manufacture fairly falling within the scope of the claims of this patent.

What is claimed is:

1. A computing system comprising:

a media device; and

a meter coupled to the media device and configured to monitor the media device, the meter comprising:

a processor; and

a non-transitory computer-readable storage medium, having stored thereon program instructions that, upon execution by the processor, cause performance of a set of operations comprising:

determining, for at least a threshold duration, that no activity was detected on a bus of an HDMI port of the media device;

injecting a data frame including an address on the bus; and

detecting a state of the media device based on a response to the data frame including the address.

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2. The computing system of claim 1, wherein the set of operations further comprising:

detecting the state of the media device to be on when the response to the data frame is received on the bus; and

detecting the state of the media device to be off when the response to the data frame is not received on the bus.

3. The computing system of claim 1, wherein the bus is a display data channel (DDC) bus of the HDMI port.

4. The computing system of claim 3, wherein the response corresponds to an acknowledgment of the data frame that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device.

5. The computing system of claim 4, wherein the set of operations further comprising:

determining the acknowledgment of the data frame is received when a voltage corresponding to an acknowledgment bit that follows the address and a read-write bit is pulled down on the DDC bus of the HDMI port.

6. The computing system of claim 1, wherein the threshold duration is a variable parameter that is configurable based on user input, a fixed value, preset value, or a compilation parameter.

7. The computing system of claim 1, wherein the media device is configured to present broadcast media, streaming media, or downloaded media.

8. The computing system of claim 1, wherein the set of operations further comprising:

providing the state of the media device to the meter.

9. A method comprising:

determining, for at least a threshold duration, that no activity was detected on a bus of an HDMI port of a media device;

injecting a data frame including an address on the bus; and detecting a state of the media device based on a response to the data frame including the address.

10. The method of claim 9, wherein the detecting of the state of the media device comprises:

detecting the state of the media device to be on when the response to the data frame is received on the bus; and detecting the state of the media device to be off when the response to the data frame is not received on the bus.

11. The method of claim 9, wherein the bus is a display data channel (DDC) bus of the HDMI port.

12. The method of claim 11, wherein the response corresponds to an acknowledgment of the data frame that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device, the I2C electrical subsystem to implement the DDC bus of the HDMI port.

13. The method of claim 12, further including determining the acknowledgment of the data frame is received when a voltage corresponding to an acknowledgment bit that follows the address and a read-write bit is pulled down on the DDC bus of the HDMI port.

14. The method of claim 9, wherein the threshold duration is a variable parameter that is configurable based on user input, a fixed value, preset value, or a compilation parameter.

15. The method of claim 9, wherein the media device is configured to present broadcast media, streaming media, or downloaded media.

16. The method of claim 9, further comprising providing the state of the media device to a meter.

17. A non-transitory computer-readable storage medium, having stored thereon program instructions that, upon execution by a processor, cause performance of a set of operations comprising:

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determining, for at least a threshold duration, that no activity was detected on a bus of an HDMI port of a media device;

injecting a data frame including an address on the bus; and detecting a state of the media device based on a response to the data frame including the address.

18. The non-transitory computer-readable storage medium of claim 17, wherein the set of operations further comprising:

detecting the state of the media device to be on when the response to the data frame is received on the bus; and detecting the state of the media device to be off when the response to the data frame is not received on the bus.

19. The non-transitory computer-readable storage medium of claim 17, wherein the bus is a display data channel (DDC) bus of the HDMI port.

20. The non-transitory computer-readable storage medium of claim 19, wherein the response corresponds to an acknowledgment of the data frame that is provided by an inter-integrated circuit (I2C) electrical subsystem of the media device.

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21. The non-transitory computer-readable storage medium of claim 20, wherein the set of operations further comprising determining the acknowledgment of the data frame is received when a voltage corresponding to an acknowledgment bit that follows the address and a read-write bit is pulled down on the DDC bus of the HDMI port.

22. The non-transitory computer-readable storage medium of claim 17, wherein the threshold duration is a variable parameter that is configurable based on user input, a fixed value, preset value, or a compilation parameter.

23. The non-transitory computer-readable storage medium of claim 17, wherein the media device is configured to present broadcast media, streaming media, or downloaded media.

24. The non-transitory computer-readable storage medium of claim 17, wherein the set of operations further comprising providing the state of the media device to a meter.

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